On the structural conjecture for extremizers of the independent Berry–Esseen inequality

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Abstract

In the proved regimes of the independent Berry–Esseen problem—including single summands, common-diameter two-point arrays, pairs of two-point summands with arbitrary diameters, and general arrays with sufficiently small Lyapunov ratio—arrays of independent centered random variables that maximize the Berry–Esseen ratio consist of $m \leq n$ identically distributed two-point random variables together with $n - m$ deterministic zero variables. In this paper we formulate the structural and value conjectures for fixed numbers of summands n, separate the determination of the optimal constant from the classification and uniqueness of maximizing arrays, and examine what current theorems and computations establish. First, we separate the choice of the marginal distribution μ of the sum $S = X_1 + \cdots + X_n$ from the choice of independent summands $(X_1, \ldots, X_n)$ that realize μ. Studying the minimum third-moment functional $\mathcal{M}_n(\mu) = \inf \sum_{i=1}^n \mathbb{E}|X_i|^3$ over n-summand independent factorizations of μ, we prove that $\mathcal{M}_n(\mu)$ is attained on a W_3^n-compact fiber and is lower semicontinuous, but for $n \geq 2$ is continuous at μ if and only if μ is convolution-indecomposable. Using an elementary third-moment splitting identity and probability generating functions, we solve the factorization problem for Poisson-binomial and binomial sum distributions and deduce that an n-summand global maximizer consists of $m \leq n$ identically distributed two-point summands and $n-m$ zeros if and only if its sum S has a centered binomial distribution of order $m \leq n$. Second, we establish a sharp two-parameter moment inequality on separated supports and an endogenous characteristic-function criterion whose zero set over all finite or countable independent arrays consists precisely of finite identically distributed two-point arrays with zero padding. Third, we synthesize the partial classifications proved in the companion manuscripts, explain what the stored high-precision binomial scans and the certified four-summand unequal-diameter local maximizer demonstrate, and isolate the missing global implication and analytical obstructions that leave the general finite-n conjectures open.

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1 Introduction: three questions on independent summands

1.1 The Berry–Esseen ratio and the padded binomial family

Let $X_1, \dots, X_n$ be independent centered real random variables with positive total variance and finite third absolute moments. Write

$$S = \sum_{i=1}^{n} X_i, \qquad V = \sum_{i=1}^{n} \mathbb{E}X_i^2 > 0, \qquad R = \sum_{i=1}^{n} \mathbb{E}|X_i|^3 < \infty, \qquad L = \frac{R}{V^{3/2}}, \tag{1.1}$$

where L is the Lyapunov ratio. Deterministic zero summands ($X_i = 0$ almost surely) are permitted; they contribute zero to S, V, and R, and merely leave available coordinate slots unused. Let

$$\Delta(S) = \sup_{x\in\mathbb{R}} \left| \mathbb{P}\left\{ \frac{S}{\sqrt{V}} \le x \right\} - \Phi(x) \right|, \qquad \mathcal{C}(X_1, \dots, X_n) = \frac{\Delta(S)}{L} = \frac{V^{3/2}\Delta(S)}{R}, \tag{1.2}$$

where $\Phi(x) = \int_{-\infty}^{x} \phi(u)\, du$ and $\phi(u) = (2\pi)^{-1/2} e^{-u^2/2}$ are the standard normal distribution function and density. Notice that $\mathcal{C}(X_1, \dots, X_n)$ is invariant under multiplying all summands by a common nonzero constant, under common sign reflection $X_i \mapsto -X_i$, and under permuting coordinates.

For each integer $n \ge 1$, let

$$C_n = \sup \mathcal{C}(X_1, \dots, X_n), \qquad C_{\mathrm{ind}} = \sup_{n\ge 1} C_n, \tag{1.3}$$

where the supremum in C_n is taken over all n-tuples of independent centered real random variables satisfying (1.1). Because an n-summand array can be extended to $n+1$ summands by adjoining $X_{n+1} = 0$, the sequence $(C_n)_{n\ge 1}$ is nondecreasing. As we recall in Section 4, the supremum C_n is

attained for every $n \geq 1$, and requiring every summand to have strictly positive variance yields the exact same supremum C_n (though the positive-variance supremum need not be attained, as occurs at $n = 2$ when the summands are restricted to two-point distributions [1, Theorem 1.2]).

A fundamental special family consists of arrays in which $m \in \{1, \ldots, n\}$ summands are identically distributed centered two-point (Bernoulli) random variables and the remaining $n - m$ summands are zero:

$$X_i = h(B_i - p) \quad (1 \leq i \leq m), \qquad X_i = 0 \quad (m < i \leq n), \tag{1.4}$$

where $h > 0$ is the common support diameter, $B_1, \ldots, B_m$ are independent $\mathrm{Bern}(p)$ random variables with $\mathbb{P}(B_i = 1) = p \in (0, 1)$ and $\mathbb{P}(B_i = 0) = q := 1 - p$. For the array (1.4),

$$V = mh^2pq, \qquad R = mh^3pq(p^2 + q^2), \qquad L = \frac{p^2 + q^2}{\sqrt{mpq}}.$$

Writing $K = \sum_{i=1}^{m} B_i \sim \mathrm{Bin}(m, p)$ and $z_k = (k - mp)/\sqrt{mpq}$, the Kolmogorov distance $\Delta(S)$ is achieved at one of the jump points $k \in \{0, 1, \ldots, m\}$ on either the right-continuous or the left-continuous branch of the distribution function, so the Berry–Esseen ratio of (1.4) is

$$R_m(p) = \frac{\sqrt{mpq}}{p^2 + q^2} \max_{0 \leq k \leq m} \{\mathbb{P}(K \leq k) - \Phi(z_k),\, \Phi(z_k) - \mathbb{P}(K < k)\}. \tag{1.5}$$

Since $R_m(p) = R_m(1 - p)$ grows like $\sqrt{mp}$ as $p \downarrow 0$, the continuous function $p \mapsto R_m(p)$ attains its maximum on $(0, 1)$. Define the m-summand binomial maximum and the n-summand padded binomial envelope by

$$B_m = \max_{0 < p < 1} R_m(p), \qquad P_n = \max_{1 \leq m \leq n} B_m. \tag{1.6}$$

Clearly $P_n \leq C_n$ for every $n \geq 1$.

In 1956, Esseen [13] proved that among identically distributed summands with finite third absolute moments, the largest asymptotic value of Δ/L as $m \to \infty$ is

$$C_{\mathrm{E}} = \frac{3 + \sqrt{10}}{6\sqrt{2\pi}} = 0.40973218370239634299\ldots, \tag{1.7}$$

and that this limit is attained by two-point summands with success probability

$$p_{\mathrm{E}} = \frac{4 - \sqrt{10}}{2} = 0.41886116991581033400\ldots \tag{1.8}$$

or $q_{\mathrm{E}} = 1 - p_{\mathrm{E}} = (\sqrt{10} - 2)/2$. Sixty years later, Schulz [24, Theorem 1] proved that $B_m < C_{\mathrm{E}}$ for every $m \geq 1$, so that $\sup_{n \geq 1} P_n = \lim_{m \to \infty} B_m = C_{\mathrm{E}}$.

1.2 Separating value, structure, and uniqueness

The central theme of this paper is the empirical and theoretical observation that arrays maximizing $\mathcal{C}(X_1, \ldots, X_n)$ appear always to belong to the padded binomial family (1.4). To analyze this phenomenon precisely, one must keep three questions distinct:

Question 1. (Optimal value) *Does $C_n = P_n$ hold for a given $n \geq 1$, and does $C_{\mathrm{ind}} = C_{\mathrm{E}}$ hold across all n?*

This is an inequality statement: it asks whether $\Delta(S) \leq P_n L$ holds for every n-summand array (and $\Delta(S) \leq C_{\mathrm{E}} L$ for all finite arrays). Because $P_n < C_{\mathrm{E}}$ for every n and $\lim_{n \to \infty} P_n = C_{\mathrm{E}}$, the equality $C_n = P_n$ for all $n \geq 1$ would immediately imply $C_{\mathrm{ind}} = C_{\mathrm{E}}$, whereas $C_{\mathrm{ind}} = C_{\mathrm{E}}$ is weaker than $C_n = P_n$ at finite n.

Question 2. (Classification of maximizing arrays) *Does every n-summand array attaining C_n have the form (1.4) up to common scaling, reflection, and coordinate permutation?*

Proving an upper bound on C_n, or even proving $C_n = P_n$, does not by itself classify all arrays that attain the maximum; conversely, knowing that every maximizer of C_n has the form (1.4) immediately implies $C_n = P_n$. Classification requires showing both that every nonzero summand is supported on two points and that all nonzero summands share the same support diameter h and the same success probability p (with the same orientation).

Question 3. (Uniqueness of the binomial parameters) *Within the padded binomial family (1.4), is the maximizing active count $m_n \in \{1, \dots, n\}$ in $P_n = \max_{1 \le m \le n} B_m$ unique, and for a given m, is the maximizing probability $p_m \in (0, 1/2]$ in $B_m = \max_{0 < p \le 1/2} R_m(p)$ unique?*

Even after an n-summand maximizer is shown to belong to the family (1.4), determining how many summands are zero and whether two distinct probabilities or two distinct active counts $m \neq m'$ can tie for the maximum P_n is a separate analytical problem in the one-variable family $R_m(p)$.

Why can a maximizer at horizon n set $n - m > 0$ coordinates to zero instead of using all n available summands? As already occurs at $n = 2$, where $B_2 \approx 0.355891 < B_1 \approx 0.370352$, and as persists for arbitrarily large n, the Kolmogorov distance of a binomial sum $\mathrm{Bin}(m, p)$ is evaluated at an integer threshold $k \in \{0, \dots, m\}$. When m is large, the optimal probability p lies within $O(m^{-1})$ of Esseen's probability p_{E}, and the optimal fractional offset $mp - k$ between the mean and the integer threshold converges to $c_* = (20 + \sqrt{10})/72 \approx 0.321698$. Because k must be an integer, $mp - k$ cannot be well aligned with c_* near $p = p_{\mathrm{E}}$ at every integer m. Since $B_m = C_{\mathrm{E}} - Q_*/m + O(m^{-2})$, reducing the active sample size from n to $m = n - j$ ($j \in \{0, 1, 2, 3, 4\}$) incurs an $O(n^{-2})$ penalty $Q_* j/n^2$ in the leading $-Q_*/m$ term while shifting the fractional lattice phase $(n - j)p_{\mathrm{E}}$ mod 1 to reduce the $O(n^{-2})$ threshold-quantization loss, strictly increasing the Berry–Esseen ratio whenever np_{E} mod 1 lies in specific phase intervals [1, Section 4].

1.3 Outline of the paper

The purpose of this paper is to give a self-contained account of the structural theory surrounding Questions 1–3, connecting the theorems proved in the seven companion manuscripts [1–7] with two exact structural mechanisms and with the repository's computational records:

- In **Section 2** we examine how choosing the marginal distribution $\mu = \mathrm{Law}(S)$ of the sum differs from choosing the independent summands $(X_1, \dots, X_n)$ that produce μ. The numerator $\Delta(S)$ and total variance V depend only on μ, so for any non-normal sum distribution μ (where $\Delta(\mu) > 0$), maximizing $\mathcal{C}(X_1, \dots, X_n)$ among independent factorizations of μ is equivalent to minimizing the third-moment sum $\sum_{i=1}^n \mathbb{E}|X_i|^3$ over all n-summand independent factorizations of μ. Using an elementary third-moment splitting identity (Proposition 2.1), we prove that the minimum third-moment functional $\mathcal{M}_n(\mu)$ is attained and lower semicontinuous in W_3, but for $n \ge 2$ is continuous at μ if and only if μ is convolution-indecomposable (Theorem 2.2). Using probability generating functions, we solve the factorization problem explicitly when μ is a Poisson-binomial or binomial law and deduce that an n-summand maximizer of C_n has the padded i.i.d. two-point form (1.4) if and only if its sum S has a centered binomial distribution of order $m \le n$ (Theorem 2.3).

- In **Section 3** we establish exact moment and characteristic-function criteria that single out padded i.i.d. two-point arrays across all finite or countable independent arrays. First, for summands whose supports have spacing at least $h > 0$, a size-biased decomposition yields a sharp two-parameter family of linear inequalities between R, hV, and the signed third moment $K_3 = \sum_i \mathbb{E}X_i^3$, with equality if and only if all nonzero summands share a single two-point law (Theorem 3.1). Second, without assuming discreteness or support separation, the moments V, R, and K_3 define a canonical scale $\widehat{h} = (R+D)/V$ and effective active count $\widehat{m} = 2V^3/(D(R+D)) \le n$ such that the characteristic-function deficit $\mathcal{J} = 1 - |\mathbb{E}e^{2\pi \mathrm{i} S/\widehat{h}}|^2$ vanishes if and only if the array is a finite i.i.d. two-point core with zero padding (Theorem 3.3). We then compare these criteria with the zero-bias Wasserstein transport inequality $2VW_1(S, S^*) \le R$ of Bell [7].

- In **Section 4** we review the quantitative bounds and partial structural classifications proved in the companion papers, showing how the general three-point variational reduction [2], the common-diameter Bernoulli classification and phase expansion [1], and the small-Lyapunov global maximizer classification [6] fit together.

- In **Section 5** we explain what the stored computational datasets actually establish, distinguishing the floating-point binomial evaluations through $m = 3000$ and compound-Poisson comparisons from the rigorous 320-bit interval certificate for a strict local maximizer at $n = 4$ with unequal support diameters $h = (1, 1, 1, 2)$.

- In **Section 6** we formulate the open finite-n value, classification, and uniqueness conjectures and discuss the specific analytical obstacles that remain between the proved regimes and the general problem.

2 Sum distributions versus independent summands

2.1 The minimum third-moment functional on a convolution fiber

In the Berry–Esseen ratio

$$\mathcal{C}(X_1, \dots, X_n) = \frac{V^{3/2}\Delta(S)}{R},$$

the total variance $V = \mathrm{Var}(S)$ and the Kolmogorov discrepancy $\Delta(S)$ depend only on the marginal distribution $\mu = \mathrm{Law}(S)$ of the sum. The individual laws $\mu_i = \mathrm{Law}(X_i)$ affect $\mathcal{C}(X_1, \dots, X_n)$ solely through the denominator $R = \sum_{i=1}^n \mathbb{E}|X_i|^3$ and the constraint that their convolution $\mu_1 * \cdots * \mu_n$ equals μ.

Let $\mathcal{P}_3^0(\mathbb{R})$ denote the space of centered Borel probability measures on $\mathbb{R}$ with finite third absolute moment, equipped with the L^3-Wasserstein distance

$$W_3(\mu, \nu) = \inf\{(\mathbb{E}|U - W|^3)^{1/3} : \mathrm{Law}(U) = \mu,\ \mathrm{Law}(W) = \nu\}.$$

For $\mu \in \mathcal{P}_3^0(\mathbb{R})$, write

$$V(\mu) = \int_{\mathbb{R}} x^2\, \mu(dx), \qquad \rho(\mu) = \int_{\mathbb{R}} |x|^3\, \mu(dx),$$

and when $V(\mu) > 0$, write

$$\Delta(\mu) = \sup_{x \in \mathbb{R}} \left| \mu((-\infty, x]) - \Phi\left(\frac{x}{\sqrt{V(\mu)}}\right) \right|.$$

For each integer $n \geq 1$ and each $\mu \in \mathcal{P}_3^0(\mathbb{R})$, define the *convolution fiber*

$$\mathcal{F}_n(\mu) = \{(\mu_1, \ldots, \mu_n) \in \mathcal{P}_3^0(\mathbb{R})^n : \mu_1 * \cdots * \mu_n = \mu\} \tag{2.1}$$

and the *minimum third-moment functional*

$$\mathcal{M}_n(\mu) = \inf_{(\mu_1,\ldots,\mu_n)\in\mathcal{F}_n(\mu)} \sum_{i=1}^n \rho(\mu_i). \tag{2.2}$$

Because the Dirac mass δ_0 belongs to $\mathcal{P}_3^0(\mathbb{R})$ with $\rho(\delta_0) = 0$, the tuple $(\mu, \delta_0, \ldots, \delta_0)$ always belongs to $\mathcal{F}_n(\mu)$, so $\mathcal{F}_n(\mu)$ is nonempty and $\mathcal{M}_n(\mu)$ is nonincreasing in n:

$$\frac{V(\mu)^{3/2}}{\sqrt{n}} \leq \mathcal{M}_n(\mu) \leq \rho(\mu). \tag{2.3}$$

Indeed, for any $(\mu_1, \ldots, \mu_n) \in \mathcal{F}_n(\mu)$ with variances $v_i = V(\mu_i)$, Lyapunov's inequality $\rho(\mu_i) \geq v_i^{3/2}$ and Hölder's inequality $\sum_{i=1}^n v_i \leq n^{1/3}\big(\sum_{i=1}^n v_i^{3/2}\big)^{2/3}$ give the lower bound in (2.3), with equality if and only if $v_1 = \cdots = v_n = V(\mu)/n$ and each μ_i is supported on $\{\pm\sqrt{V(\mu)/n}\}$ with equal probabilities $1/2$.

Decomposing the supremum in (1.3) first over the sum distribution $\mu = \mathrm{Law}(S)$ and then over its convolution fiber $\mathcal{F}_n(\mu)$ gives

$$C_n = \sup_{\substack{\mu\in\mathcal{P}_3^0(\mathbb{R})\\ V(\mu)>0}} \frac{V(\mu)^{3/2}\Delta(\mu)}{\mathcal{M}_n(\mu)}. \tag{2.4}$$

In particular, if $(X_1, \ldots, X_n)$ is any array attaining C_n with sum distribution $\mu = \mathrm{Law}(X_1 + \cdots + X_n)$, then since $C_n \geq C_1 > 0$ implies $\Delta(\mu) > 0$, the summand distributions $\mu_i = \mathrm{Law}(X_i)$ must minimize $\sum_{i=1}^n \rho(\mu_i)$ over $\mathcal{F}_n(\mu)$:

$$\sum_{i=1}^n \mathbb{E}|X_i|^3 = \mathcal{M}_n(\mu). \tag{2.5}$$

2.2 The third-moment splitting identity

To understand which factorizations minimize $\sum_{i=1}^n \rho(\mu_i)$ on a fixed fiber $\mathcal{F}_n(\mu)$, the basic tool is the strict superadditivity of the third absolute moment under convolution of independent centered variables [10, 23, Theorem (2.6)]. The following elementary identity makes the exact convolution excess completely explicit without requiring symmetry or decomposition into two-point mixtures.

Proposition 2.1 (Third-moment splitting identity)**.** *Let X and Y be independent centered real random variables with finite third absolute moments, and set $m = \min(|X|, |Y|)$ and $M = \max(|X|, |Y|)$. Then*

$$\mathbb{E}|X+Y|^3 - \mathbb{E}|X|^3 - \mathbb{E}|Y|^3 = 2\,\mathbb{E}\big[\mathbf{1}_{\{XY<0\}} m^2(3M - m)\big] \geq 0. \tag{2.6}$$

Moreover, equality holds in (2.6) if and only if at least one of X and Y is zero almost surely.

Proof. We first verify that for all real numbers $x, y \in \mathbb{R}$,

$$|x+y|^3 - |x|^3 - |y|^3 - 3xy(|x| + |y|) = 2\mathbf{1}_{\{xy<0\}} \min(|x|, |y|)^2\big(3\max(|x|, |y|) - \min(|x|, |y|)\big). \tag{2.7}$$

If $xy \geq 0$, then $|x+y| = |x| + |y|$, so $|x+y|^3 - |x|^3 - |y|^3 = 3|x|\,|y|(|x| + |y|) = 3xy(|x| + |y|)$, and both sides of (2.7) vanish. If $xy < 0$, then by symmetry we may assume $a = |x| \geq b = |y| > 0$, so that $xy = -ab$ and $|x+y| = a - b$; the left-hand side of (2.7) is then

$$(a-b)^3 - a^3 - b^3 + 3ab(a+b) = 2b^2(3a - b),$$

which coincides with the right-hand side of (2.7).

Since $\mathbb{E}|X|^3 < \infty$ and $\mathbb{E}|Y|^3 < \infty$, every term in (2.7) is integrable when evaluated at (X, Y). Because X and Y are independent and centered,

$$\mathbb{E}[XY|X|] = \mathbb{E}[X|X|]\,\mathbb{E}Y = 0, \qquad \mathbb{E}[XY|Y|] = \mathbb{E}X\,\mathbb{E}[Y|Y|] = 0.$$

Taking expectations in (2.7) therefore yields (2.6).

On the event $\{XY < 0\}$ we have $M \geq m > 0$, so $m^2(3M - m) \geq 2m^3 > 0$. If neither X nor Y is zero almost surely, then since $\mathbb{E}X = \mathbb{E}Y = 0$, each of X and Y takes both strictly positive and strictly negative values with positive probability; by independence, $\mathbb{P}(XY < 0) = \mathbb{P}(X > 0)\mathbb{P}(Y < 0) + \mathbb{P}(X < 0)\mathbb{P}(Y > 0) > 0$, and therefore the right-hand side of (2.6) is strictly positive. □

2.3 Compactness, attainment, and discontinuity of the fiber minimum

A measure $\mu \in \mathcal{P}_3^0(\mathbb{R})$ is called *convolution-indecomposable* (or simply *indecomposable* in $\mathcal{P}_3^0(\mathbb{R})$) if every factorization $\mu = \mu_1 * \mu_2$ with $\mu_1, \mu_2 \in \mathcal{P}_3^0(\mathbb{R})$ has $\mu_1 = \delta_0$ or $\mu_2 = \delta_0$. By Proposition 2.1, splitting a decomposable summand into two nonconstant independent centered factors preserves the distribution of the sum S while strictly decreasing the sum of third absolute moments. This strict decrease has a striking consequence for the regularity of $\mathcal{M}_n(\mu)$ as a function of the sum law μ: although $\mathcal{M}_n$ is attained on every fiber and is lower semicontinuous, it is discontinuous at every decomposable distribution whenever $n \geq 2$.

Theorem 2.2 (Compactness, attainment, and continuity of $\mathcal{M}_n$)**.** *Fix an integer $n \geq 1$.*

(i) For every W_3-compact subset $\mathcal{K} \subset \mathcal{P}_3^0(\mathbb{R})$, the union of convolution fibers $\bigcup_{\mu \in \mathcal{K}} \mathcal{F}_n(\mu)$ is compact in $\mathcal{P}_3^0(\mathbb{R})^n$ under the product W_3 topology. In particular, for every $\mu \in \mathcal{P}_3^0(\mathbb{R})$, the fiber $\mathcal{F}_n(\mu)$ is W_3^n-compact, the infimum $\mathcal{M}_n(\mu)$ in (2.2) is attained, and $\mu \mapsto \mathcal{M}_n(\mu)$ is lower semicontinuous on $(\mathcal{P}_3^0(\mathbb{R}), W_3)$.

(ii) If $n \geq 2$, then for every $\mu \in \mathcal{P}_3^0(\mathbb{R})$,

$$\mathcal{M}_n(\mu) = \rho(\mu) \quad \Longleftrightarrow \quad \mu \text{ is convolution-indecomposable.} \tag{2.8}$$

(iii) Convolution-indecomposable finitely supported measures are dense in $(\mathcal{P}_3^0(\mathbb{R}), W_3)$. Consequently, for fixed $n \geq 2$, the functional $\mathcal{M}_n$ is continuous at $\mu \in \mathcal{P}_3^0(\mathbb{R})$ if and only if μ is convolution-indecomposable.

(iv) On the open domain $\{\mu \in \mathcal{P}_3^0(\mathbb{R}) : V(\mu) > 0\}$, the optimized Berry–Esseen ratio

$$\mathcal{Q}_n(\mu) = \frac{V(\mu)^{3/2}\Delta(\mu)}{\mathcal{M}_n(\mu)}$$

is upper semicontinuous in W_3. For fixed $n \geq 2$, $\mathcal{Q}_n$ is continuous at μ if and only if either μ is convolution-indecomposable or μ is a centered normal distribution $N(0, V)$.

Proof. (i) Let $(X_1, \dots, X_n)$ be independent centered random variables with $\mathrm{Law}(X_i) = \mu_i$ and $\mathrm{Law}(S) = \mu \in \mathcal{K}$, where $S = X_1 + \cdots + X_n$. Because the summands are independent and centered, $\mathbb{E}[S \mid X_i] = X_i$ almost surely for each $i \in \{1, \dots, n\}$. For every $A \geq 0$, the function $f_A(x) = (|x| - A)_+^3$ is convex on $\mathbb{R}$, and $|x|^3 \mathbf{1}_{\{|x|>2A\}} \leq 8 f_A(x)$ since $|x| > 2A$ implies $|x| - A > |x|/2$. Conditional Jensen's inequality therefore gives

$$\int_{\{|x|>2A\}} |x|^3 \, \mu_i(dx) = \mathbb{E}\big[|X_i|^3 \mathbf{1}_{\{|X_i|>2A\}}\big] \leq 8 \, \mathbb{E} f_A(X_i) \leq 8 \, \mathbb{E} f_A(S) = 8 \int_{\{|x|>A\}} (|x| - A)^3 \, \mu(dx). \tag{2.9}$$

Since $\mathcal{K}$ is W_3-compact, its third absolute moments are bounded and uniformly integrable, so the right-hand side of (2.9) tends to 0 as $A \to \infty$ uniformly over $\mu \in \mathcal{K}$. Hence each marginal coordinate projection of $\bigcup_{\mu \in \mathcal{K}} \mathcal{F}_n(\mu)$ is tight with uniformly integrable third absolute moments, and is therefore relatively compact in $(\mathcal{P}_3^0(\mathbb{R}), W_3)$. Because convergence in W_3 preserves zero mean and convolution $(\mu_1, \dots, \mu_n) \mapsto \mu_1 * \cdots * \mu_n$ is continuous in W_3 (by coupling each coordinate independently and using Minkowski's inequality), $\bigcup_{\mu \in \mathcal{K}} \mathcal{F}_n(\mu)$ is closed and hence compact in W_3^n. Taking $\mathcal{K} = \{\mu\}$ shows that $\mathcal{F}_n(\mu)$ is compact, and since $(\mu_1, \dots, \mu_n) \mapsto \sum_{i=1}^n \rho(\mu_i)$ is continuous on W_3^n, the infimum $\mathcal{M}_n(\mu)$ is attained. If $\mu^{(k)} \to \mu$ in W_3, applying compactness to the W_3-compact set $\mathcal{K} = \{\mu\} \cup \{\mu^{(k)} : k \geq 1\}$ and extracting a convergent subsequence of attaining tuples $(\mu_1^{(k)}, \dots, \mu_n^{(k)}) \in \mathcal{F}_n(\mu^{(k)})$ yields $\mathcal{M}_n(\mu) \leq \liminf_{k \to \infty} \mathcal{M}_n(\mu^{(k)})$.

(ii) Let $n \geq 2$. If $\mu = \nu_1 * \nu_2$ with $\nu_1, \nu_2 \in \mathcal{P}_3^0(\mathbb{R}) \setminus \{\delta_0\}$, then $(\nu_1, \nu_2, \delta_0, \dots, \delta_0) \in \mathcal{F}_n(\mu)$ has third-moment sum $\rho(\nu_1) + \rho(\nu_2) < \rho(\mu)$ by Proposition 2.1, so $\mathcal{M}_n(\mu) < \rho(\mu)$. Conversely, if $(\mu_1, \dots, \mu_n) \in \mathcal{F}_n(\mu)$ has at least two nondegenerate components, say $\mu_1 \neq \delta_0$ and $\mu_2 \neq \delta_0$, then grouping $\nu_1 = \mu_1$ and $\nu_2 = \mu_2 * \cdots * \mu_n$ gives a nontrivial factorization $\mu = \nu_1 * \nu_2$ (since $\mathrm{Var}(\nu_2) = \sum_{i=2}^n \mathrm{Var}(\mu_i) \geq \mathrm{Var}(\mu_2) > 0$), so μ is decomposable. Thus when μ is indecomposable, every tuple in $\mathcal{F}_n(\mu)$ has at most one nonzero component, which must equal μ, and hence $\mathcal{M}_n(\mu) = \rho(\mu)$.

(iii) Suppose $\nu \in \mathcal{P}_3^0(\mathbb{R})$ is supported on a finite set $S = \{x_1 < \cdots < x_s\}$ with $\nu(\{x_j\}) > 0$ for all j, and suppose that all $\binom{s}{2}$ positive differences $x_r - x_j$ $(1 \leq j < r \leq s)$ are distinct. We claim that ν is convolution-indecomposable. Indeed, if $\nu = \nu_1 * \nu_2$ with neither factor degenerate, then $\mathrm{supp}(\nu_1)$ contains at least two points $a_0 < a_1$ and $\mathrm{supp}(\nu_2)$ contains at least two points $b_0 < b_1$. Because all four sums $a_0 + b_0$, $a_1 + b_0$, $a_0 + b_1$, $a_1 + b_1$ belong to $\mathrm{supp}(\nu) = S$, either $a_1 + b_0 = a_0 + b_1$ (in which case the three distinct points $a_0 + b_0 < a_1 + b_0 < a_1 + b_1$ of S repeat the positive difference $a_1 - a_0 = b_1 - b_0$) or the four sums are four distinct points of S among which the positive difference $a_1 - a_0 = (a_1 + b_0) - (a_0 + b_0) = (a_1 + b_1) - (a_0 + b_1)$ occurs twice, contradicting the assumption that all positive differences in S are distinct.

Any $\mu \in \mathcal{P}_3^0(\mathbb{R})$ can be approximated arbitrarily closely in W_3 by finitely supported centered measures, and an arbitrarily small perturbation of the support points outside the finite union of hyperplanes where two coordinates or two pairwise differences coincide (followed by subtracting the perturbed mean, which shifts all support points by the same constant and preserves all pairwise differences) produces a sequence $\mu^{(k)} \to \mu$ in W_3 of finitely supported indecomposable measures. For fixed $n \geq 2$, if μ is indecomposable, then for any sequence $\nu^{(k)} \to \mu$ in W_3, lower semicontinuity and the upper bound $\mathcal{M}_n(\nu^{(k)}) \leq \rho(\nu^{(k)}) \to \rho(\mu) = \mathcal{M}_n(\mu)$ force $\mathcal{M}_n(\nu^{(k)}) \to \mathcal{M}_n(\mu)$. Conversely, if μ is decomposable, then $\mathcal{M}_n(\mu) < \rho(\mu)$ by (ii), whereas along a sequence of indecomposable measures $\mu^{(k)} \to \mu$ in W_3 we have $\mathcal{M}_n(\mu^{(k)}) = \rho(\mu^{(k)}) \to \rho(\mu) > \mathcal{M}_n(\mu)$, so $\mathcal{M}_n$ is discontinuous at μ.

(iv) For any two centered probability measures μ, ν with finite first moments and any coupling (U, W) with $\mathbb{E}|U - W| = W_1(\mu, \nu)$, centering gives $\mathbb{E}(W - U)_+ = \mathbb{E}(U - W)_+ = \frac{1}{2}\mathbb{E}|U - W| =$

$\frac{1}{2}W_1(\mu,\nu)$. By Markov's inequality and the bound $\|\Phi'\|_\infty = \phi(0)$, for every $x \in \mathbb{R}$ and $\delta > 0$,

$$\mathbb{P}(U \le x) - \Phi(x) \le \mathbb{P}(W \le x+\delta) - \Phi(x+\delta) + \phi(0)\delta + \frac{\mathbb{E}(W-U)_+}{\delta}.$$

Choosing $\delta = \sqrt{W_1(\mu,\nu)/(2\phi(0))}$ and repeating the argument for the opposite sign and with U and W interchanged yields

$$\left| \sup_{x\in\mathbb{R}} |\mathbb{P}(U \le x) - \Phi(x)| - \sup_{x\in\mathbb{R}} |\mathbb{P}(W \le x) - \Phi(x)| \right| \le \sqrt{2\phi(0) W_1(\mu,\nu)}.$$

Since standardization $\mu \mapsto \mathrm{Law}(S/\sqrt{V(\mu)})$ is continuous in W_3 on $\{V(\mu) > 0\}$, the numerator $V(\mu)^{3/2}\Delta(\mu)$ is continuous in W_3. Because $\mathcal{M}_n(\mu) \ge V(\mu)^{3/2}/\sqrt{n} > 0$ is lower semicontinuous by (i), the quotient $\mathcal{Q}_n(\mu)$ is upper semicontinuous.

If μ is indecomposable, both numerator and denominator of $\mathcal{Q}_n$ are continuous at μ. If $\mu = N(0,V)$, then $\Delta(\mu) = 0$, and as $\nu \to \mu$ in W_3 the bound $\mathcal{M}_n(\nu) \ge V(\nu)^{3/2}/\sqrt{n}$ gives $0 \le \mathcal{Q}_n(\nu) \le \sqrt{n}\,\Delta(\nu) \to 0 = \mathcal{Q}_n(\mu)$, so $\mathcal{Q}_n$ is continuous at μ even though $\mathcal{M}_n$ jumps from $\rho(N(0,V))$ along indecomposable approximants to $\mathcal{M}_n(N(0,V)) = \rho(N(0,V))/\sqrt{n}$. Finally, if μ is decomposable and not normal, then $\Delta(\mu) > 0$ while $\mathcal{M}_n(\mu) < \rho(\mu)$; along indecomposable approximants $\mu^{(k)} \to \mu$ in W_3,

$$\lim_{k\to\infty} \mathcal{Q}_n(\mu^{(k)}) = \frac{V(\mu)^{3/2}\Delta(\mu)}{\rho(\mu)} < \frac{V(\mu)^{3/2}\Delta(\mu)}{\mathcal{M}_n(\mu)} = \mathcal{Q}_n(\mu),$$

so $\mathcal{Q}_n$ is discontinuous at μ. □

Theorem 2.2 explains why the reformulation (2.4) over marginal sum laws μ cannot be treated as a smooth variational problem on $\mathcal{P}_3^0(\mathbb{R})$: the denominator $\mathcal{M}_n(\mu)$ drops downward precisely on the thin locus of distributions μ that admit nontrivial independent factorizations.

2.4 Factorizations of Poisson-binomial and binomial sum distributions

When the sum distribution μ is a centered Poisson-binomial or binomial distribution, we can determine all elements of the convolution fiber $\mathcal{F}_n(\mu)$ explicitly and identify the unique factorizations that achieve $\mathcal{M}_n(\mu)$.

Theorem 2.3 (Factorizations of Poisson-binomial and binomial laws)**.** *Fix an integer $n \ge 1$ and a lattice span $g > 0$.*

(i) Let $\mu = \mathrm{Law}(g\sum_{j=1}^k (B_j - p_j))$, where $B_1, \dots, B_k$ are independent with $B_j \sim \mathrm{Bern}(p_j)$ and $0 < p_j < 1$. Then $(\mu_1, \dots, \mu_n) \in \mathcal{F}_n(\mu)$ if and only if there exists a partition of $\{1, \dots, k\}$ into n (possibly empty) disjoint subsets $A_1, \dots, A_n$ such that

$$\mu_i = \mathrm{Law}\left(g \sum_{j\in A_i} (B_j - p_j) \right) \qquad (1 \le i \le n), \tag{2.10}$$

where an empty sum denotes 0. If $n \ge k$, then $\mathcal{M}_n(\mu)$ is equal to

$$\mathcal{M}_n(\mu) = g^3 \sum_{j=1}^k p_j(1-p_j)(p_j^2 + (1-p_j)^2), \tag{2.11}$$

and the minimum is attained if and only if $(\mu_1, \dots, \mu_n)$ is a permutation of the k singleton two-point laws $\mathrm{Law}(g(B_j - p_j))$ $(1 \le j \le k)$ together with $n-k$ Dirac masses at 0.

(ii) In particular, for any n-summand independent array $(X_1, \dots, X_n)$ that attains the n-summand supremum C_n, the following two statements are equivalent:

- *(a) $(X_1, \dots, X_n)$ consists of $m \in \{1, \dots, n\}$ identically distributed nondegenerate two-point summands and $n - m$ zero summands;*
- *(b) the marginal distribution of the sum $S = X_1 + \cdots + X_n$ is a centered binomial distribution $\mathrm{Law}(g(\mathrm{Bin}(m, p) - mp))$ for some integer $m \in \{1, \dots, n\}$, some $g > 0$, and some $p \in (0, 1)$.*

(iii) For each integer $N \geq 1$, let $\mu_N = \mathrm{Law}\big((\mathrm{Bin}(N, 1/2) - N/2)/\sqrt{N/4}\big)$ denote the standardized symmetric binomial distribution of order N. Then for every capacity $n \geq 1$, every oversized symmetric binomial sum law ($N > n$) has strictly smaller optimized ratio $\mathcal{Q}_n(\mu_N) = \Delta(\mu_N)/\mathcal{M}_n(\mu_N)$ than μ_n, and

$$\sup_{N \geq 1} \mathcal{Q}_n(\mu_N) = \begin{cases} R_1(1/2) = \Phi(1) - 1/2 & \text{if } n = 1, \\ \max\{R_n(1/2),\, R_{n-1}(1/2)\} & \text{if } n \geq 2. \end{cases} \tag{2.12}$$

Proof. (i) Let $(X_1, \dots, X_n)$ be independent centered random variables with $\mathrm{Law}(X_i) = \mu_i$ and $\mathrm{Law}(X_1 + \cdots + X_n) = \mu$. The support of μ is the finite arithmetic progression $-g\sum_{j=1}^{k} p_j + g\{0, 1, \dots, k\}$. Fixing one point in $\mathrm{supp}(X_r)$ for each $r \neq i$, the sum of any point in $\mathrm{supp}(X_i)$ with those fixed points lies in $\mathrm{supp}(\mu)$ because the X_r are independent. Consequently, $\mathrm{supp}(X_i)$ is finite and all differences between points of $\mathrm{supp}(X_i)$ belong to $g\mathbb{Z}$. Let $a_i = \min \mathrm{supp}(X_i)$; then $Y_i = (X_i - a_i)/g$ takes nonnegative integer values with $\mathbb{P}(Y_i = 0) > 0$, and independence gives $\sum_{i=1}^{n} a_i = \min \mathrm{supp}(\mu) = -g\sum_{j=1}^{k} p_j$ and $\mathrm{Law}(Y_1 + \cdots + Y_n) = \mathrm{Law}(B_1 + \cdots + B_k)$.

Let $P_i(z) = \mathbb{E}z^{Y_i}$ $(1 \leq i \leq n)$ be the probability generating polynomials of $Y_1, \dots, Y_n$. Then

$$\prod_{i=1}^{n} P_i(z) = \prod_{j=1}^{k} (1 - p_j + p_j z) \qquad (z \in \mathbb{C}).$$

By unique factorization in $\mathbb{C}[z]$ and the normalization $P_i(1) = 1$, each polynomial $P_i(z)$ is the product of $(1 - p_j + p_j z)$ over a subset $A_i \subseteq \{1, \dots, k\}$, where $A_1, \dots, A_n$ form a partition of $\{1, \dots, k\}$. Since $\mathbb{E}X_i = 0$ determines the translation $a_i = -g\sum_{j \in A_i} p_j$, the representation (2.10) follows.

Now suppose $n \geq k$. By repeated application of Proposition 2.1, for every subset $A_i \subseteq \{1, \dots, k\}$,

$$\mathbb{E}\left| g \sum_{j \in A_i} (B_j - p_j) \right|^3 \geq g^3 \sum_{j \in A_i} \mathbb{E}|B_j - p_j|^3,$$

with equality if and only if $|A_i| \leq 1$. Summing over $i = 1, \dots, n$ yields (2.11), with equality if and only if every A_i has cardinality 0 or 1—that is, k of the subsets A_i are singletons and the remaining $n - k$ are empty.

(ii) Implication (a) $\Longrightarrow$ (b) is immediate from the definition of the binomial distribution. Conversely, suppose $(X_1, \dots, X_n)$ attains C_n and its sum S has distribution $\mu = \mathrm{Law}(g(\mathrm{Bin}(m, p) - mp))$ with $1 \leq m \leq n$, $g > 0$, and $0 < p < 1$. By (2.5), the summand laws $\mu_i = \mathrm{Law}(X_i)$ must achieve the minimum $\mathcal{M}_n(\mu)$ on $\mathcal{F}_n(\mu)$. Applying part (i) with $k = m$ and $p_1 = \cdots = p_m = p$ shows that the unique minimizers of $\sum_{i=1}^{n} \rho(\mu_i)$ on $\mathcal{F}_n(\mu)$ up to coordinate permutation are m copies of $\mathrm{Law}(g(B_1 - p))$ and $n - m$ copies of δ_0, which is (a).

(iii) By Mattner and Schulz [20, Theorem 1.3(a)], for the standardized symmetric binomial law μ_N $(p = 1/2)$, the maximum Kolmogorov discrepancy $d_N := \Delta(\mu_N)$ is attained at the central

thresholds $\lfloor N/2 \rfloor$ (on the right-continuous branch) and $\lceil N/2 \rceil$ (on the left-continuous branch), and equals

$$d_{2k} = \frac{1}{2}\binom{2k}{k}2^{-2k}, \qquad d_{2k+1} = \Phi\Big(\frac{1}{\sqrt{2k+1}}\Big) - \frac{1}{2} = \int_0^{1/\sqrt{2k+1}} \phi(u)\,du. \tag{2.13}$$

We first verify that d_N is strictly decreasing in $N \geq 1$: $d_1 > d_2 > d_3 > \cdots$. Writing $u_k = \binom{2k}{k}2^{-2k}$ and $I_r = \int_0^{\pi/2} \sin^r \theta\, d\theta$, integration by parts gives $I_{2k} = (\pi/2)u_k$, $I_{2k+1} = 1/((2k+1)u_k)$, and $I_{2k-1} = 1/(2ku_k)$; since $I_{2k+1} < I_{2k} < I_{2k-1}$ for all $k \geq 1$, Wallis's inequalities

$$\frac{1}{\sqrt{\pi(k+1/2)}} < u_k < \frac{1}{\sqrt{\pi k}}$$

hold for all $k \geq 1$. Since $\phi(0) = 1/\sqrt{2\pi}$, the left inequality gives

$$d_{2k} = \frac{u_k}{2} > \frac{\phi(0)}{\sqrt{2k+1}} > \int_0^{1/\sqrt{2k+1}} \phi(u)\,du = d_{2k+1}.$$

On the other hand, for every odd $r = 2k-1 \geq 1$, the inequality $e^{-u^2/2} \geq 1 - u^2/2$ and the upper Wallis bound give

$$d_r = \int_0^{1/\sqrt{r}} \phi(u)\,du \geq \frac{\phi(0)}{\sqrt{r}}\left(1 - \frac{1}{6r}\right) > \frac{\phi(0)}{\sqrt{r+1}} > \frac{u_k}{2} = d_{r+1},$$

because $(1+1/r)(1-1/(6r))^2 - 1 = (24r^2 - 11r + 1)/(36r^3) > 0$ for $r \geq 1$. Thus $d_N > d_{N+1}$ for all $N \geq 1$.

Now let $N > n$. By (2.3), $\mathcal{M}_n(\mu_N) \geq 1/\sqrt{n}$, whereas for $N = n$ part (i) gives $\mathcal{M}_n(\mu_n) = 1/\sqrt{n}$. Since $d_N < d_n$, we obtain

$$\mathcal{Q}_n(\mu_N) = \frac{d_N}{\mathcal{M}_n(\mu_N)} \leq \sqrt{n}\,d_N < \sqrt{n}\,d_n = \frac{d_n}{\mathcal{M}_n(\mu_n)} = R_n(1/2).$$

For $1 \leq N \leq n$, part (i) gives $\mathcal{M}_n(\mu_N) = 1/\sqrt{N}$, so $\mathcal{Q}_n(\mu_N) = \sqrt{N}\,d_N = R_N(1/2)$. When $n = 1$ this leaves only $R_1(1/2)$. When $n \geq 2$, within each parity $R_N(1/2)$ is strictly increasing in N: for odd N, $R_N(1/2) = \sqrt{N}\int_0^{1/\sqrt{N}} \phi(u)\,du = \int_0^1 \phi(v/\sqrt{N})\,dv$ increases strictly with N, and for even $N = 2k$, $R_{2k+2}(1/2)/R_{2k}(1/2) = (2k+1)/(2\sqrt{k(k+1)}) > 1$. Therefore $\max_{1\leq N\leq n} R_N(1/2) = \max\{R_n(1/2), R_{n-1}(1/2)\}$, proving (2.12). □

Remark 2.4 (Balanced groupings for general $N > n$, and Gaussian and Poisson fibers). When $\mu = \mathrm{Law}(g(\mathrm{Bin}(N,p) - Np))$ with $N > n$ and $0 < p < 1$, Theorem 2.3(i) shows that every factorization in $\mathcal{F}_n(\mu)$ has the form $\mu_i = \mathrm{Law}(g(\mathrm{Bin}(k_i,p) - k_i p))$ with nonnegative integers $k_1 + \cdots + k_n = N$. Writing $c_p(k) = \mathbb{E}|\,\mathrm{Bin}(k,p) - kp|^3$ and $U \sim \mathrm{Bern}(p) - p$, the identity $\frac{d^2}{dx^2}(\mathbb{E}|x+U|^3 - |x|^3) = 6(\mathbb{E}|x+U| - |x|) \geq 0$ and conditional Jensen's inequality imply that $k \mapsto c_p(k+1) - c_p(k)$ is nondecreasing, so c_p is convex on $\mathbb{Z}_{\geq 0}$ and $\mathcal{M}_n(\mu)$ is attained by balanced group sizes $k_i \in \{\lfloor N/n \rfloor, \lceil N/n \rceil\}$.

By contrast, for infinitely divisible sum laws μ, the fiber minimum $\mathcal{M}_n(\mu)$ behaves quite differently from the finite-lattice case:

(1) If $\mu = N(0,V)$, then by Cramér's theorem [11] every factor in $\mathcal{F}_n(\mu)$ is normal $N(0,v_i)$ with $\sum_{i=1}^n v_i = V$, and by strict convexity of $v \mapsto v^{3/2}$ the unique minimizer of $\sum_{i=1}^n \rho(\mu_i) = \rho(N(0,1))\sum_{i=1}^n v_i^{3/2}$ consists of n identically distributed $N(0,V/n)$ summands (with $\mathcal{M}_n(N(0,V)) = \rho(N(0,V))/\sqrt{n}$).

(2) If $\mu = \mathrm{Law}(N_T - T)$ with $N_T \sim \mathrm{Pois}(T)$ ($T > 0$), then for any factorization $(X_1, \dots, X_n)$ of $N_T - T$ the signed third moments add ($\sum_{i=1}^n \mathbb{E}X_i^3 = \mathbb{E}(N_T - T)^3 = T$), so

$$\sum_{i=1}^n \mathbb{E}|X_i|^3 - T = \sum_{i=1}^n (\mathbb{E}|X_i|^3 - \mathbb{E}X_i^3) = 2\sum_{i=1}^n \mathbb{E}(X_i)_-^3 > 0,$$

because every nonconstant centered variable takes strictly negative values with positive probability. On the other hand, taking n independent copies of $N_{T/n} - T/n$ gives, for $n \geq T$,

$$\mathcal{M}_n(\mu) \leq n\,\mathbb{E}|N_{T/n} - T/n|^3 = T + \frac{2T^3}{n^2} e^{-T/n} \longrightarrow T \qquad (n \to \infty),$$

so $\mathcal{M}_n(\mu) > T$ for every finite n while $\inf_{n\geq 1} \mathcal{M}_n(\mu) = T$. (By Raikov's theorem [22], every factor in $\mathcal{F}_n(\mu)$ is a centered Poisson law $N_{t_i} - t_i$ with $\sum_{i=1}^n t_i = T$.) Notice that the limiting quantity $T = \int y^3\,\nu(dy)$ is the third moment of the Lévy jump measure $\nu = T\delta_1$ arising from $n \to \infty$ rare summands, whereas at any fixed finite n every independent factorization of $N_T - T$ has third-moment sum strictly greater than T.

Theorem 2.3(ii) clarifies what the structural classification question (Question 2) actually requires: because exact-fiber third-moment minimization forces every third-moment-minimizing factorization of $g(\mathrm{Bin}(m,p) - mp)$ ($m \leq n$) in $\mathcal{F}_n(\mu)$ to consist of m identically distributed two-point summands and $n - m$ zeros, proving that an n-summand maximizer of C_n has the padded i.i.d. two-point form (1.4) is equivalent to proving that the marginal distribution of its sum S is a centered binomial distribution of order $m \leq n$.

3 Moment, transport, and characteristic-function criteria for identical two-point summands

3.1 The separated-support moment inequality

Why do identically distributed two-point summands emerge when one compares Kolmogorov discrepancy with the third absolute moment sum R? In his 1956 construction, Esseen [13, Lemma 1, pp. 162–163] proved that for any purely atomic random variable X whose support points are separated by at least $h > 0$,

$$3h\,\mathrm{Var}(X) + |\mathbb{E}(X - \mathbb{E}X)^3| \leq (3 + \sqrt{10})\,\mathbb{E}|X - \mathbb{E}X|^3, \tag{3.1}$$

with equality for nonconstant X if and only if $X - \mathbb{E}X$ is a two-point random variable of support span h and upper probability p_{E} or $1 - p_{\mathrm{E}}$. By decomposing a centered distribution into its size-biased positive and negative parts, we obtain an exact two-parameter family of sum-of-squares identities containing (3.1) that classifies all two-point equality laws of arbitrary skewness.

Theorem 3.1 (Separated-support moment inequality and equality cases)**.** *(i) Let X be a nonconstant centered real random variable with finite third absolute moment whose support* $\mathrm{supp}(X)$ *satisfies $|x - y| \geq h > 0$ for all distinct $x, y \in \mathrm{supp}(X)$. Set*

$$v = \mathbb{E}X^2, \qquad \kappa = \mathbb{E}X^3, \qquad M_+ = \mathbb{E}(X_+)^3, \qquad M_- = \mathbb{E}(X_-)^3, \qquad \rho = \mathbb{E}|X|^3 = M_+ + M_-,$$

where $X_+ = \max(X, 0)$ and $X_- = \max(-X, 0)$. Then

$$hv \leq \left(\sqrt{M_+} + \sqrt{M_-}\right)^2 = \rho + 2\sqrt{M_+ M_-}, \tag{3.2}$$

and for every $\lambda > 0$ and $\nu \in \mathbb{R}$, setting $d = \sqrt{\lambda^2 + \nu^2}$,

$$(\lambda + d)\rho - \lambda h v - \nu\kappa = \lambda\Big(\big(\sqrt{M_+} + \sqrt{M_-}\big)^2 - hv\Big) + \Big(\sqrt{d-\nu}\sqrt{M_+} - \sqrt{d+\nu}\sqrt{M_-}\Big)^2 \geq 0. \tag{3.3}$$

Equality holds in (3.3) if and only if X is supported on two points $\{-a, b\}$ with $a, b > 0$, $a + b = h$, and upper atom probability

$$p = \mathbb{P}(X = b) = \frac{a}{h} = \frac{1}{2}\left(1 - \frac{\nu}{\lambda + d}\right). \tag{3.4}$$

(ii) Let $(X_i)_{i\in I}$ be a finite or countable collection of independent centered real random variables with $0 < V = \sum_{i\in I} \mathbb{E}X_i^2 < \infty$ and $R = \sum_{i\in I} \mathbb{E}|X_i|^3 < \infty$, and suppose that for every nonconstant X_i the points of $\operatorname{supp}(X_i)$ are separated by at least the same $h > 0$. Set $K_3 = \sum_{i\in I} \mathbb{E}X_i^3$. Then for every $\lambda > 0$ and $\nu \in \mathbb{R}$ with $d = \sqrt{\lambda^2 + \nu^2}$,

$$(\lambda + d)R - \lambda h V - \nu K_3 \geq 0, \tag{3.5}$$

with equality if and only if $(X_i)_{i\in I}$ consists of finitely many identically distributed two-point random variables of support span h and upper atom probability (3.4), together with deterministic zero variables.

Proof. (i) Because X is centered and nonconstant, $m := \mathbb{E}X_+ = \mathbb{E}X_- > 0$. Define positive random variables B and A with distributions

$$\mathbb{P}(B \in E) = \frac{1}{m}\mathbb{E}\big[X_+ \mathbf{1}_{\{X\in E\cap(0,\infty)\}}\big], \qquad \mathbb{P}(A \in E) = \frac{1}{m}\mathbb{E}\big[X_- \mathbf{1}_{\{-X\in E\cap(0,\infty)\}}\big].$$

Then $v = \mathbb{E}X_+^2 + \mathbb{E}X_-^2 = m(\mathbb{E}B + \mathbb{E}A)$, $M_+ = m\,\mathbb{E}B^2$, and $M_- = m\,\mathbb{E}A^2$. For every $b \in \operatorname{supp}(B) \subset \operatorname{supp}(X) \cap (0,\infty)$ and $a \in \operatorname{supp}(A) \subset (-\operatorname{supp}(X)) \cap (0,\infty)$, the points $b > 0$ and $-a < 0$ are distinct elements of $\operatorname{supp}(X)$, so the separation hypothesis gives $a + b \geq h$. Hence $\mathbb{E}A + \mathbb{E}B \geq h$, and the Cauchy–Schwarz inequality $\mathbb{E}B \leq \sqrt{\mathbb{E}B^2}$, $\mathbb{E}A \leq \sqrt{\mathbb{E}A^2}$ yields

$$hv \leq m(\mathbb{E}A + \mathbb{E}B)^2 \leq m\Big(\sqrt{\mathbb{E}A^2} + \sqrt{\mathbb{E}B^2}\Big)^2 = \big(\sqrt{M_-} + \sqrt{M_+}\big)^2,$$

which is (3.2). Expanding the two terms on the right-hand side of (3.3) and using $\rho = M_+ + M_-$, $\kappa = M_+ - M_-$, and $\sqrt{(d-\nu)(d+\nu)} = \lambda$ gives the identity (3.3).

Both terms on the right-hand side of (3.3) are nonnegative. The first term vanishes if and only if $\mathbb{E}A + \mathbb{E}B = h$, $\operatorname{Var}(A) = 0$, and $\operatorname{Var}(B) = 0$, which means that $A = a > 0$ and $B = b > 0$ are deterministic constants with $a + b = h$. In that case $\operatorname{supp}(X) \setminus \{0\} = \{-a, b\}$; since $0 < a, b < h$, the point 0 has distance $\min(a, b) < h$ to $\{-a, b\}$ and therefore cannot belong to $\operatorname{supp}(X)$ under the hypothesis that all points of $\operatorname{supp}(X)$ are separated by at least h. Thus $\operatorname{supp}(X) = \{-a, b\}$, and $\mathbb{E}X = 0$ forces $p = \mathbb{P}(X = b) = a/h$ and $q = \mathbb{P}(X = -a) = b/h$, with $M_+ = pb^3 = h^3pq^3$ and $M_- = qa^3 = h^3p^3q$. The second term on the right-hand side of (3.3) vanishes if and only if

$$\frac{q}{p} = \frac{b}{a} = \sqrt{\frac{M_+}{M_-}} = \sqrt{\frac{d+\nu}{d-\nu}} = \frac{d+\nu}{\lambda},$$

which is equivalent to $q - p = (d + \nu - \lambda)/(d + \nu + \lambda) = \nu/(\lambda + d)$, giving (3.4).

(ii) Summing (3.3) over all nonconstant summands X_i (while zero summands contribute 0 to R, V, and K_3) gives (3.5). Equality holds if and only if every nonconstant summand achieves equality in (3.3) with the same parameters (h, λ, ν), and hence has the same two-point distribution of span h and upper atom probability (3.4). Since each such nonconstant summand has variance $h^2pq > 0$ and $V = \sum_{i\in I} \mathbb{E}X_i^2 < \infty$, only finitely many summands can be nonconstant. □

Remark 3.2 (Classical specializations). Setting $\lambda = 3$ and $\nu = \pm 1$ in Theorem 3.1 gives $d = \sqrt{10}$, $\lambda + d = 3 + \sqrt{10}$, and upper atom probability $p = \frac{1}{2}(1 \mp 1/(3 + \sqrt{10})) \in \{p_{\mathrm{E}}, 1 - p_{\mathrm{E}}\}$, recovering Esseen's inequality (3.1) (which governs the leading Edgeworth–Euler–Maclaurin term for half-line probabilities $\mathbb{P}(S \le x)$). Setting $\lambda = 1$ and $\nu = 0$ gives $d = 1$, $\lambda + d = 2$, and $hv \le 2\rho$, with equality uniquely at symmetric two-point laws ($p = 1/2$); this is the moment inequality governing the asymptotic Berry–Esseen constant $\sqrt{2/\pi}$ for interval probabilities $\mathbb{P}(a < S \le b)$ established by Dinev and Mattner [12, Theorem 2.1 and Corollary 2.3]. Note also that the assumption in Theorem 3.1(i) that *all* support points of X are separated by at least h (and not merely that opposite-signed support points satisfy $a + b \ge h$) is necessary for the two-point equality conclusion: if an atom at 0 with probability $r \in (0, 1)$ is permitted at distance less than h from $\{-a, b\}$, then the centered three-point distribution assigning probabilities $(1-r)b/h$ to $-a$, r to 0, and $(1-r)a/h$ to b (where $a, b > 0$, $a + b = h$, and $b/a = (d + \nu)/\lambda$) also makes the right-hand side of (3.3) vanish.

3.2 An endogenous characteristic-function criterion

Theorem 3.1 assumes a common lower bound $h > 0$ on the support spacings of the summands. Remarkably, the moments of an arbitrary independent array determine a canonical scale $\widehat{h}$ and a single characteristic-function value whose modulus equals 1 if and only if the array is a finite i.i.d. two-point core with zero padding—without any a priori discreteness, lattice, or support-separation assumption.

Theorem 3.3 (Characteristic-function criterion and moment capacity bound)**.** *Let $(X_i)_{i \in I}$ be a finite or countable collection of independent centered real random variables with $0 < V = \sum_{i \in I} \mathbb{E}X_i^2 < \infty$ and $R = \sum_{i \in I} \mathbb{E}|X_i|^3 < \infty$, and let $S = \sum_{i \in I} X_i$ (converging in L^2 and almost surely). Set*

$$\begin{aligned} A &= \sum_{i \in I} \mathbb{E}(X_{i,+})^3, & B &= \sum_{i \in I} \mathbb{E}(X_{i,-})^3, \\ K_3 &= A - B = \sum_{i \in I} \mathbb{E}X_i^3, & D &= 2\sqrt{AB} = \sqrt{R^2 - K_3^2} > 0, \end{aligned} \tag{3.6}$$

and define the moment-determined span, skewness, effective active count, and characteristic-function deficit by

$$\begin{aligned} \widehat{h} &= \frac{R + D}{V}, & \widehat{s} &= \frac{K_3}{R + D} \in (-1, 1), \\ \widehat{m} &= \frac{2V^3}{D(R + D)}, & \mathcal{J} &= 1 - \left| \mathbb{E} \exp\left(\frac{2\pi \mathrm{i} S}{\widehat{h}} \right) \right|^2 \ge 0. \end{aligned} \tag{3.7}$$

Then:

(i) If the index set $I = \{1, \dots, n\}$ is finite, then

$$0 < \widehat{m} \le n, \tag{3.8}$$

with equality $\widehat{m} = n$ if and only if all n summands $X_1, \dots, X_n$ are identically distributed nondegenerate two-point random variables.

(ii) For any finite or countable index set I,

$$\mathcal{J} = 0 \iff \begin{array}{l} (X_i)_{i \in I} \text{ consists of finitely many i.i.d.} \\ \text{nondegenerate two-point summands and zeros.} \end{array} \tag{3.9}$$

When $\mathcal{J} = 0$, the number of nondegenerate summands equals $\widehat{m} \in \mathbb{N}$, their common support diameter is $\widehat{h}$, and their common upper atom probability is $p = (1 - \widehat{s})/2$.

(iii) Fix a finite horizon $n \geq 1$ and $L_0 < \infty$. If $(X_1^{(k)}, \dots, X_n^{(k)})$ is a sequence of n-summand independent centered arrays with $V^{(k)} = 1$, $R^{(k)} \leq L_0$, and $\mathcal{J}^{(k)} \to 0$ as $k \to \infty$, then every subsequence has a further subsequence converging coordinatewise in W_3 to an array consisting of $m \in \{1, \dots, n\}$ identically distributed nondegenerate two-point summands and $n - m$ zeros.

Proof. First note that since $V > 0$ and $\mathbb{E}X_i = 0$, at least one X_i is nonconstant and takes both positive and negative values with positive probability, so $A > 0$, $B > 0$, and $D = 2\sqrt{AB} > 0$.

(i) For a single nonconstant centered summand X_i, write $v_i = \mathbb{E}X_i^2$, $A_i = \mathbb{E}(X_{i,+})^3$, $B_i = \mathbb{E}(X_{i,-})^3$, $\rho_i = A_i + B_i$, $\delta_i = 2\sqrt{A_i B_i}$, and $m_i = \mathbb{E}X_{i,+} = \mathbb{E}X_{i,-} > 0$. By Hölder's inequality on $\{X_i > 0\}$ and $\{X_i < 0\}$,

$$m_i = \mathbb{E}[X_{i,+}\mathbf{1}_{\{X_i>0\}}] \leq A_i^{1/3}\mathbb{P}(X_i > 0)^{2/3}, \qquad m_i = \mathbb{E}[X_{i,-}\mathbf{1}_{\{X_i<0\}}] \leq B_i^{1/3}\mathbb{P}(X_i < 0)^{2/3}.$$

Since $\mathbb{P}(X_i > 0) + \mathbb{P}(X_i < 0) \leq 1$, summing the $3/2$-powers gives

$$m_i^{3/2}\left(\frac{1}{\sqrt{A_i}} + \frac{1}{\sqrt{B_i}}\right) \leq 1, \qquad \text{or} \qquad m_i \leq \left(\frac{\sqrt{A_i B_i}}{\sqrt{A_i} + \sqrt{B_i}}\right)^{2/3}.$$

Meanwhile, Cauchy–Schwarz gives $v_i = \mathbb{E}X_{i,+}^2 + \mathbb{E}X_{i,-}^2 \leq \sqrt{m_i}(\sqrt{A_i} + \sqrt{B_i})$. Raising to the power $3/2$ and substituting the bound on m_i yields

$$v_i^{3/2} \leq m_i^{3/4}(\sqrt{A_i} + \sqrt{B_i})^{3/2} \leq \sqrt{\sqrt{A_i B_i}}\,(\sqrt{A_i} + \sqrt{B_i}) = \sqrt{\frac{\delta_i(\rho_i + \delta_i)}{2}}, \tag{3.10}$$

which also holds trivially when $X_i = 0$. Equality holds in (3.10) for $v_i > 0$ if and only if $\mathbb{P}(X_i = 0) = 0$ and $X_{i,+}$, $X_{i,-}$ are constant on $\{X_i > 0\}$ and $\{X_i < 0\}$, meaning that X_i is a two-point random variable. By Cauchy–Schwarz and Minkowski's inequality $\sum_{i=1}^n \delta_i = 2\sum_{i=1}^n \sqrt{A_i B_i} \leq 2\sqrt{AB} = D$, summing (3.10) over $i = 1, \dots, n$ gives

$$\frac{V^{3/2}}{\sqrt{n}} \leq \sum_{i=1}^n v_i^{3/2} \leq \sqrt{\frac{1}{2}\left(\sum_{i=1}^n \delta_i\right)\left(R + \sum_{i=1}^n \delta_i\right)} \leq \sqrt{\frac{D(R+D)}{2}},$$

which rearranges to $\widehat{m} = 2V^3/(D(R+D)) \leq n$. Equality $\widehat{m} = n$ requires $v_1 = \cdots = v_n = V/n > 0$, each X_i two-point, and (A_i, B_i) proportional across all i, which forces all n two-point summands to be identically distributed.

(ii) If $(X_i)_{i\in I}$ consists of $m \geq 1$ identically distributed two-point summands $g(B_i - p)$ ($g > 0$, $0 < p < 1$, $q = 1 - p$) and zeros, then $V = mg^2pq$, $A = mg^3pq^3$, $B = mg^3p^3q$, $R = mg^3pq(p^2 + q^2)$, $K_3 = mg^3pq(q - p)$, and $D = 2mg^3p^2q^2$, so $R + D = mg^3pq(p + q)^2 = mg^3pq$. Hence $\widehat{h} = (R + D)/V = g$, $\widehat{s} = q - p$, $\widehat{m} = m$, and since every summand is supported on a coset of $g\mathbb{Z}$, $\left|\mathbb{E}e^{2\pi \mathrm{i}S/g}\right| = \prod_{i\in I}\left|\mathbb{E}e^{2\pi \mathrm{i}X_i/g}\right| = 1$, so $\mathcal{J} = 0$.

Conversely, suppose $\mathcal{J} = 0$. By independence,

$$1 = \left|\mathbb{E}\exp\left(\frac{2\pi \mathrm{i}S}{\widehat{h}}\right)\right|^2 = \prod_{i\in I}\left|\mathbb{E}\exp\left(\frac{2\pi \mathrm{i}X_i}{\widehat{h}}\right)\right|^2,$$

so $\left|\mathbb{E}\exp(2\pi \mathrm{i}X_i/\widehat{h})\right| = 1$ for every $i \in I$. Thus for each $i \in I$, $\exp(2\pi \mathrm{i}X_i/\widehat{h})$ is almost surely constant, so $\mathrm{supp}(X_i)$ is contained in a single coset of $\widehat{h}\mathbb{Z}$. In particular, for every nonconstant X_i, the points of $\mathrm{supp}(X_i)$ are separated by at least $\widehat{h}$. Applying Theorem 3.1(ii) with span $h = \widehat{h}$,

$\lambda = 1$, and $\nu = K_3/D$, we have $d = \sqrt{1 + K_3^2/D^2} = R/D$ and $\lambda + d = (R+D)/D$, so the left-hand side of (3.5) equals

$$\frac{R+D}{D}R - \frac{R+D}{V}V - \frac{K_3}{D}K_3 = \frac{R(R+D) - K_3^2}{D} - (R+D) = \frac{R^2 - K_3^2 + RD}{D} - (R+D) = 0,$$

because $R^2 - K_3^2 = D^2$. Therefore equality holds in (3.5), and Theorem 3.1(ii) implies that $(X_i)_{i\in I}$ consists of finitely many identically distributed two-point summands of span $\widehat{h}$ and upper probability $p = \frac{1}{2}(1 - \nu/(\lambda + d)) = (1 - \widehat{s})/2$, together with zeros.

(iii) Let $(X_1^{(k)}, \dots, X_n^{(k)})$ have $V^{(k)} = 1$, $R^{(k)} \le L_0$, and $\mathcal{J}^{(k)} \to 0$. Because $\sum_{i=1}^n \mathbb{E}|X_i^{(k)}|^3 \le L_0$, the sequence of n-tuples is tight in $\mathbb{R}^n$ with uniformly integrable second moments. Passing to a subsequence, we may assume that $X_i^{(k)} \Rightarrow X_i^{(\infty)}$ weakly with $\mathbb{E}X_i^{(\infty)} = 0$ and $\sum_{i=1}^n \mathbb{E}(X_i^{(\infty)})^2 = 1$, while $A_i^{(k)} = \mathbb{E}(X_{i,+}^{(k)})^3 \to a_i$ and $B_i^{(k)} = \mathbb{E}(X_{i,-}^{(k)})^3 \to b_i$. By Fatou's lemma for weak convergence, $A_i' := \mathbb{E}(X_{i,+}^{(\infty)})^3 \le a_i$ and $B_i' := \mathbb{E}(X_{i,-}^{(\infty)})^3 \le b_i$. Let $A' = \sum_{i=1}^n A_i' > 0$, $B' = \sum_{i=1}^n B_i' > 0$, $a = \sum_{i=1}^n a_i \ge A'$, and $b = \sum_{i=1}^n b_i \ge B'$. Then

$$\widehat{h}^{(k)} = \big(\sqrt{\textstyle\sum_i A_i^{(k)}} + \sqrt{\textstyle\sum_i B_i^{(k)}}\big)^2 \longrightarrow H := \big(\sqrt{a} + \sqrt{b}\big)^2 \in [n^{-1/2}, 2L_0].$$

Since $|\mathbb{E}e^{itX_i^{(k)}} - \mathbb{E}e^{iuX_i^{(k)}}| \le |t-u|\,\mathbb{E}|X_i^{(k)}| \le |t-u|$, weak convergence $X_i^{(k)} \Rightarrow X_i^{(\infty)}$ and $\widehat{h}^{(k)} \to H > 0$ imply $|\mathbb{E}e^{2\pi i S^{(\infty)}/H}|^2 = \lim_{k\to\infty}(1 - \mathcal{J}^{(k)}) = 1$. Thus each nonconstant limiting summand $X_i^{(\infty)}$ is supported on a coset of $H\mathbb{Z}$ and has support spacing at least H. Summing (3.2) of Theorem 3.1 over the nonconstant limiting summands (with total variance 1) and using Minkowski's inequality gives

$$H = H\sum_{i=1}^n \mathbb{E}(X_i^{(\infty)})^2 \le \sum_{i=1}^n \Big(\sqrt{A_i'} + \sqrt{B_i'}\Big)^2 \le \Big(\sqrt{A'} + \sqrt{B'}\Big)^2 \le \Big(\sqrt{a} + \sqrt{b}\Big)^2 = H.$$

Hence every inequality in this chain is an equality: $A' = a$, $B' = b$, and so $A_i' = a_i$ and $B_i' = b_i$ for each $i = 1, \dots, n$. Thus $\mathbb{E}|X_i^{(k)}|^3 \to \mathbb{E}|X_i^{(\infty)}|^3$ alongside weak convergence, which implies $W_3(X_i^{(k)}, X_i^{(\infty)}) \to 0$, and $\mathcal{J}^{(\infty)} = 0$ together with part (ii) shows that $(X_1^{(\infty)}, \dots, X_n^{(\infty)})$ is a padded i.i.d. two-point array. □

Remark 3.4 (Diagnostic equality criteria versus objective monotonicity). Theorem 3.3 characterizes the exact zero set $\mathcal{J} = 0$ across the entire domain of independent arrays, without any prior restriction to discrete or common-diameter distributions. However, $\mathcal{J} = 0$ is a structural equality criterion rather than a monotonicity statement for the Berry–Esseen ratio $\mathcal{C}$: neither Theorem 3.1 nor Theorem 3.3 proves that a global maximizer of $\mathcal{C}$ must satisfy $\mathcal{J} = 0$, or that decreasing $\mathcal{J}$ increases $\mathcal{C}$. As we discuss in Section 6, proving the structural conjecture by this route requires supplying the missing global implication—namely, showing that every n-summand array with $\mathcal{J} > 0$ admits an admissible replacement with strictly larger Berry–Esseen ratio $\mathcal{C}$.

3.3 Comparison with zero-bias Wasserstein transport

A third global characterization of padded i.i.d. two-point arrays comes from Stein's method and the zero-bias transformation [15]. For a centered random variable X with variance $v = \mathbb{E}X^2 > 0$, its *zero-biased distribution* X^* is characterized by the identity $\mathbb{E}[Xf(X)] = v\,\mathbb{E}f'(X^*)$ for all absolutely continuous functions $f : \mathbb{R} \to \mathbb{R}$ with bounded derivative. For a finite independent array

$(X_1, \dots, X_n)$ with sum S, total variance $V > 0$, and third-moment sum $R < \infty$, the L^1-Wasserstein distance $W_1(S, S^*) = \int_{\mathbb{R}} |\mathbb{P}(S \le x) - \mathbb{P}(S^* \le x)|\, dx$ satisfies the normalized transport inequality

$$\mathcal{B}(X_1, \dots, X_n) := \frac{2VW_1(S, S^*)}{R} \le 1 \tag{3.11}$$

[14, 27]. In the companion paper Bell [7, Theorem 1.1], it is proved via Choquet's theorem on fixed-variance moment sets and a conditional-variance identity that

$$\mathcal{B}(X_1, \dots, X_n) = 1 \quad \Longleftrightarrow \quad \begin{array}{l}(X_1, \dots, X_n) \text{ consists of i.i.d.} \\ \text{nondegenerate two-point summands and zeros.}\end{array} \tag{3.12}$$

Furthermore, let $S_1, S_2, \dots$ be independent copies of the block sum $S = X_1 + \cdots + X_n$, and define the m-fold repeated-block Berry–Esseen ratio (which is distinct from the unrestricted m-summand supremum C_m) by

$$\begin{aligned} \mathcal{C}_{\mathrm{rep},m}(X_1, \dots, X_n) &:= \frac{(mV)^{3/2}\Delta(S_1 + \cdots + S_m)}{mR} \\ &= \frac{\sqrt{m}\, V^{3/2}}{R} \sup_{x \in \mathbb{R}} \left| \mathbb{P}\left\{ \frac{S_1 + \cdots + S_m}{\sqrt{mV}} \le x \right\} - \Phi(x) \right|. \end{aligned} \tag{3.13}$$

Combining (3.12) with the sharp lattice transport inequality

$$|\mathbb{E}X^3| + 3h \operatorname{Var}(X) \le (3 + \sqrt{10})\, 2 \operatorname{Var}(X) W_1(X, X^*)$$

for variables supported on a coset of $h\mathbb{Z}$ proves that for any fixed finite block $(X_1, \dots, X_n)$,

$$\lim_{m \to \infty} \mathcal{C}_{\mathrm{rep},m}(X_1, \dots, X_n) \le C_{\mathrm{E}}\, \mathcal{B}(X_1, \dots, X_n) \le C_{\mathrm{E}}, \tag{3.14}$$

with equality in (3.14) if and only if all nonconstant summands in the block share a single Esseen two-point law with probability p_{E} or $1 - p_{\mathrm{E}}$ [7, Theorem 1.2]. Along any sequence of independent arrays with $L_n \to 0$ and $\mathcal{C}_n \to C_{\mathrm{E}}$, one likewise has $\mathcal{B}_n \to 1$ [6, Theorem 1.1].

One might ask whether the asymptotic bound (3.14) holds pointwise for finite arrays—that is, whether $\mathcal{C}(X_1, \dots, X_n) \le C_{\mathrm{E}}\, \mathcal{B}(X_1, \dots, X_n)$ holds for all independent arrays. The following explicit one-summand family shows that no constant $K < \infty$ can satisfy $\mathcal{C}(X) \le K\, \mathcal{B}(X)$ for all centered distributions X.

Proposition 3.5 (Failure of pointwise domination by zero-bias transport)**.** *For each $a > 0$, let X_a be the centered real random variable that agrees with $Z \sim N(0,1)$ on $\{|x| > a\}$ and replaces the restriction of Z to $[-a, a]$ by three atoms at $\{-a, 0, a\}$ having the same total mass, zero mean, and second moment as $Z\mathbf{1}_{\{|Z| \le a\}}$. Then $\mathbb{E}X_a = 0$, $\operatorname{Var}(X_a) = 1$, and as $a \downarrow 0$,*

$$\mathcal{C}(X_a) = \frac{a}{6} + O(a^3), \qquad \mathcal{B}(X_a) = \frac{5a^2}{18} + O(a^4), \qquad \frac{\mathcal{C}(X_a)}{\mathcal{B}(X_a)} = \frac{3}{5a} + O(a) \longrightarrow \infty. \tag{3.15}$$

Proof. Let $m_a = \int_{-a}^{a} \phi(u)\, du = 2a\phi(0) + O(a^3)$ and $v_a = \int_{-a}^{a} u^2 \phi(u)\, du = \frac{2}{3}a^3\phi(0) + O(a^5)$. To match the mass, zero mean, and second moment of $Z\mathbf{1}_{\{|Z| \le a\}}$ on $\{-a, 0, a\}$, X_a places mass $w_a = v_a/(2a^2) = \frac{1}{3}a\phi(0) + O(a^3)$ at $-a$ and at $+a$, and mass $m_a - 2w_a = \frac{4}{3}a\phi(0) + O(a^3)$ at 0. Then $\operatorname{Var}(X_a) = \operatorname{Var}(Z) = 1$, and

$$R_a = \mathbb{E}|X_a|^3 = \mathbb{E}|Z|^3 + 2a^3 w_a - \int_{-a}^{a} |u|^3 \phi(u)\, du = 4\phi(0) + O(a^4).$$

Outside $(-a,a)$, the distribution function $F_{X_a}(x)$ coincides with $\Phi(x)$. On $(-a,0)$ it is constant and equal to $\Phi(-a)+w_a$, while on $[0,a)$ it equals $\Phi(a)-w_a$. Thus the maximum of $|F_{X_a}(x)-\Phi(x)|$ is attained as $x\uparrow 0$ (and by symmetry at $x=0$):

$$\Delta(X_a)=\Phi(0)-(\Phi(-a)+w_a)=\frac{1}{2}m_a-w_a=\frac{2}{3}a\phi(0)+O(a^3),$$

giving $\mathcal{C}(X_a)=\Delta(X_a)/R_a=a/6+O(a^3)$.

On the other hand, because $\mathrm{Var}(X_a)=1$, the zero-biased distribution X_a^* has Lebesgue density $f_{X_a^*}(x)=\mathbb{E}[X_a\mathbf{1}_{\{X_a>x\}}]$. For $|x|>a$, $\mathbb{E}[X_a\mathbf{1}_{\{|X_a|\le a\}}]=0$ implies $f_{X_a^*}(x)=\int_x^\infty u\phi(u)\,du=\phi(x)$. For $x\in(-a,0)\cup(0,a)$,

$$f_{X_a^*}(x)=\int_a^\infty u\phi(u)\,du+aw_a=\phi(a)+\frac{v_a}{2a}.$$

Consequently, $F_{X_a^*}(x)=\Phi(x)$ for $|x|\ge a$, and on $[-a,a]$ the distribution function $F_{X_a^*}$ is affine with slope $\phi(a)+v_a/(2a)=\phi(0)+O(a^2)$, interpolating from $\Phi(-a)$ at $-a$ to $\Phi(a)$ at a. Hence

$$\begin{aligned}W_1(X_a,X_a^*)&=\int_{-a}^{a}\left|F_{X_a}(x)-F_{X_a^*}(x)\right|dx\\&=2\int_{-a}^{0}\left|w_a-(x+a)(\phi(0)+O(a^2))\right|dx\\&=2\phi(0)\int_0^a\left|\frac{a}{3}-u\right|du+O(a^4)=2\phi(0)\left(\frac{a^2}{18}+\frac{4a^2}{18}\right)+O(a^4)=\frac{5}{9}a^2\phi(0)+O(a^4).\end{aligned}$$

Dividing $2W_1(X_a,X_a^*)$ by $R_a=4\phi(0)+O(a^4)$ gives $\mathcal{B}(X_a)=5a^2/18+O(a^4)$ and completes the proof of (3.15). □

4 Known value bounds and partial classifications of maximizers

4.1 Attainment of C_n and equivalence with positive-variance arrays

Before reviewing the known upper bounds and structural classifications, we record the basic existence theorem for fixed-n maximizers (see also 6, Lemma 2.1) and verify that permitting deterministic zero summands does not enlarge the n-summand supremum C_n.

Proposition 4.1 (Attainment of C_n and positive-variance approximation)**.** *For every integer $n\ge 1$:*

(i) The supremum $C_n=\sup\mathcal{C}(X_1,\dots,X_n)$ over n-tuples of independent centered real random variables with $V>0$ and $R<\infty$ (permitting zero summands) is attained, and every maximizing sequence normalized to $V=1$ has a subsequence converging coordinatewise in W_3 to a maximizing array.

(ii) The supremum of $\mathcal{C}(X_1,\dots,X_n)$ over n-tuples of independent centered real random variables satisfying $\mathrm{Var}(X_i)>0$ and $\mathbb{E}|X_i|^3<\infty$ for every $i\in\{1,\dots,n\}$ equals C_n.

Proof. (i) Let $(X_1^{(k)},\dots,X_n^{(k)})$ be a maximizing sequence for C_n. By scale invariance we may assume $V^{(k)}=\sum_{i=1}^n\mathbb{E}(X_i^{(k)})^2=1$. Since $C_n\ge C_1>0$ and $\Delta(S^{(k)})\le 1$, for all large k we have

$$n^{-1/2}\le R^{(k)}=\frac{\Delta(S^{(k)})}{\mathcal{C}(X_1^{(k)},\dots,X_n^{(k)})}\le\frac{2}{C_1}<\infty.$$

For each coordinate $i \in \{1, \dots, n\}$ and any $M > 0$, Markov's inequality gives

$$\mathbb{E}\big[(X_i^{(k)})^2 \mathbf{1}_{\{|X_i^{(k)}|>M\}}\big] \le \frac{R^{(k)}}{M} \le \frac{2}{MC_1}.$$

Hence the laws of $X_i^{(k)}$ are tight with uniformly integrable first and second moments. Passing to a subsequence, we may assume that $X_i^{(k)} \Rightarrow X_i^{(\infty)}$ weakly as $k \to \infty$ for each $i = 1, \dots, n$, with $\mathbb{E}X_i^{(\infty)} = 0$, $V^{(\infty)} = \sum_{i=1}^n \mathbb{E}(X_i^{(\infty)})^2 = 1$, and, by Fatou's lemma,

$$\mathbb{E}|X_i^{(\infty)}|^3 \le \liminf_{k\to\infty} \mathbb{E}|X_i^{(k)}|^3 \qquad (1 \le i \le n).$$

By independence, the sums $S^{(k)} = \sum_{i=1}^n X_i^{(k)}$ converge weakly and in W_1 to $S^{(\infty)} = \sum_{i=1}^n X_i^{(\infty)}$. Because the normal distribution function Φ is continuous with $\|\Phi'\|_\infty = \phi(0)$, the estimate $\big|\Delta(S^{(k)}) - \Delta(S^{(\infty)})\big| \le \sqrt{2\phi(0) W_1(S^{(k)}, S^{(\infty)})}$ established in the proof of Theorem 2.2(iv) (which includes both closed and strict traces at any atoms of $S^{(\infty)}$) implies that $\Delta(S^{(k)}) \to \Delta(S^{(\infty)}) = C_n \limsup_{k\to\infty} R^{(k)} > 0$. Consequently,

$$C_n = \frac{\Delta(S^{(\infty)})}{\limsup_{k\to\infty} R^{(k)}} \le \frac{\Delta(S^{(\infty)})}{\sum_{i=1}^n \mathbb{E}|X_i^{(\infty)}|^3} \le C_n.$$

Both inequalities must be equalities, so $(X_1^{(\infty)}, \dots, X_n^{(\infty)})$ attains C_n and $\lim_{k\to\infty} \mathbb{E}|X_i^{(k)}|^3 = \mathbb{E}|X_i^{(\infty)}|^3$ for every $i = 1, \dots, n$, which together with weak convergence gives $W_3(X_i^{(k)}, X_i^{(\infty)}) \to 0$.

(ii) Let $(X_1, \dots, X_n)$ be an array attaining C_n with $V = 1$. For each $\varepsilon > 0$, define $X_i^{(\varepsilon)} = X_i$ if $\mathrm{Var}(X_i) > 0$ and let $X_i^{(\varepsilon)} \sim N(0, \varepsilon)$ (independent of all other coordinates) if $X_i = 0$. Then $\mathrm{Var}(X_i^{(\varepsilon)}) > 0$ for all $i = 1, \dots, n$, while as $\varepsilon \downarrow 0$ we have $V^{(\varepsilon)} \to 1$, $R^{(\varepsilon)} = R + O(\varepsilon^{3/2}) \to R$, and $\Delta(S^{(\varepsilon)}) \to \Delta(S)$, so $\mathcal{C}(X_1^{(\varepsilon)}, \dots, X_n^{(\varepsilon)}) \to C_n$. □

Allowing zero summands is therefore the natural closed compactification of the fixed-n domain: as Bell [1, Theorem 1.2] shows for $n = 2$ two-point summands, the supremum over strictly positive-variance two-point arrays equals C_1 because one summand's variance can degenerate to 0, and that supremum is attained only when the zero summand is included in the domain.

4.2 Quantitative upper bounds on C_n and C_{ind}

We next summarize the quantitative upper bounds on C_n and C_{ind} established in the literature and in Manuscripts 1–3 [3–5]:

(1) *One summand ($n = 1$):* Bentkus and Kirša [8, Theorem 1] proved that for a single centered random variable X_1,

$$C_1 = B_1 = \max_{0<p<1} \frac{\sqrt{p(1-p)}}{p^2 + (1-p)^2} \left\{ \Phi\left(\sqrt{\frac{p}{1-p}} \right) - p \right\} = 0.370351523474378920\ldots < C_{\mathrm{E}}, \quad (4.1)$$

attained up to scale and sign reflection only by the two-point distribution at $p_1 = 0.390958631281332815\ldots$.

(2) *Two arbitrary summands ($n = 2$):* Bell [3, Theorem 1.1] proves that over all pairs of independent centered random variables,

$$C_2 < \frac{49}{120} = 0.408333\ldots < C_{\mathrm{E}}. \quad (4.2)$$

The proof fixes the threshold and individual variances, uses strong quadratic moment duality [9, 28] to reduce each summand to at most three support points, and rules out all combinatorial contact configurations above 49/120 by variational identities and certified interval enclosures.

(3) *Two-point arrays and variance-concentrated arrays:* For n identically distributed two-point summands, Schulz [24, Theorem 1] proved the strict binomial bound $B_n < C_{\mathrm{E}}$. Bell [3, Theorems 1.2 and 1.3] extend the sharp inequality $\Delta(S) \le C_{\mathrm{E}} L$ to two broader classes of non-identically distributed two-point arrays:

   (a) arbitrary numbers $n \ge 1$ of independent two-point summands whose nondegenerate members share a common support diameter $h > 0$;

   (b) at most three independent two-point summands ($n \le 3$) with arbitrary support diameters (with the uniform strict gap $C_{\mathrm{E}} L - \Delta(S) > 3/80$ when $n \le 2$).

   Combining these bounds with a Stein–Ornstein–Uhlenbeck normal comparison for the remaining summands yields $\Delta(S) \le C_{\mathrm{E}} L$ for any finite independent array in which two two-point summands carry at least 22/23 of the total variance V, or two arbitrary summands carry at least 439/440 of V [3, Corollary 1.4].

(4) *General independent arrays with small Lyapunov ratio:* Adapting the large-sample i.i.d. method of He and Cheng [16] to non-identically distributed summands, Bell [5, Theorem 1.1] proves that for every finite independent array,

$$L \le \exp(-\exp(97)) \quad \Longrightarrow \quad \Delta(S) \le C_{\mathrm{E}} L. \tag{4.3}$$

   If a counterexample existed below this threshold, maximizing Δ/L minus a smooth localization penalty (bounded using the asymptotic smoothing inequality $\Delta \le C_{\mathrm{E}} L + 0.2538 L^{4/3}$ of 26, Corollary 4.18) would produce an attained extremal array with $|X_i| \le 8L\sqrt{V}$; uniform jitter smoothing, quantitative stability in Esseen's moment inequality (3.1), and conditional Fourier and clipping comparisons against a common-diameter two-point array (bounded by $C_{\mathrm{E}} L$ via 3, Theorem 1.2(ii)) then rule out any violation.

(5) *Unconditional upper bound across all independent arrays:* Bell [4, Theorem 1.1] proves that for every $n \ge 1$ and every independent array,

$$C_{\mathrm{ind}} \le U_* < 0.44988, \tag{4.4}$$

   improving upon the bound 0.5583 of Shevtsova [25] by combining a zero-bias characteristic-function differential inequality, a signed cosine-product reference [21], isolation of the largest-variance summand, and outward-rounded interval subdivision over the moment parameter space.

4.3 General variational support reductions and compactification

What can be proved about maximizing arrays without restricting either the support of the summands or the size of the Lyapunov ratio L? In Manuscript 7 [2], variational perturbations of a single summand's distribution—fixing the maximizing threshold z and the deleted sum $T_i = \sum_{j \ne i} X_j$—establish two general necessary conditions:

(1) *Four-point and three-point support reduction* [2, Theorem 3.2]: At any local maximum of $\mathcal{C}$ with respect to two-sided probability perturbations of X_i preserving total mass 1 and mean

0 (while allowing $v_i = \mathbb{E}X_i^2$ and $\rho_i = \mathbb{E}|X_i|^3$ to vary), the second logarithmic derivative of $\mathcal{C}$ restricts the linear span of the moment vectors $(1, u, u^2, |u|^3)$ on $\mathrm{supp}(X_i)$, proving that every summand of a maximizer is supported on at most four points. At a global maximum, any four-point summand lies on a flat probability segment whose endpoint replaces X_i by a three-point summand with the exact same variance v_i and third absolute moment ρ_i (and the same ratio $\mathcal{C}$).

(2) *Convolution indecomposability when a spare slot is available* [2, Corollary 3.4]: By the third-moment splitting identity (Proposition 2.1), every nonconstant summand X_i of an unrestricted-horizon maximizer—or of a fixed-n maximizer that has at least one zero summand—is convolution-indecomposable, since splitting a decomposable summand into two nonconstant independent centered factors preserves the sum S and total variance V while strictly decreasing R. In particular, if such a maximizer has a three-point summand supported on an arithmetic progression $\{x_0, x_0+g, x_0+2g\}$ $(g>0)$ with probabilities (m_0, m_1, m_2), then $m_1^2 < 4m_0m_2$ (equivalently, $\mathrm{Var}(X_i) > ab/2$ when the centered endpoints are $-a < 0 < b$), since otherwise its probability generating polynomial factors into two nondegenerate two-point laws of diameter g.

When studying the universal constant $C_{\mathrm{ind}} = \sup_{n\geq 1} C_n$, if $C_{\mathrm{ind}} > C_{\mathrm{E}}$ then the small-Lyapunov bound (4.3) forces every unit-variance maximizing sequence to satisfy $R^{(k)} > \exp(-\exp(97))$. Passing to the limit along variances ordered nonincreasingly produces an attained maximizer in a Lévy–Khintchine compactification [2, Definition 1.1 and Theorem 1.1]:

$$\begin{aligned} S &= \sum_{i\geq 1} X_i + Y, \qquad \sum_{i\geq 1} \mathrm{Var}(X_i) + a + \int_{\mathbb{R}\setminus\{0\}} y^2\,\nu(dy) = 1, \\ \beta &= \sum_{i\geq 1} \mathbb{E}|X_i|^3 + \int_{\mathbb{R}\setminus\{0\}} |y|^3\,\nu(dy), \end{aligned} \tag{4.5}$$

where $(X_i)_{i\geq 1}$ are independent centered discrete summands (each reducible to at most three support points), Y is an independent centered infinitely divisible random variable with Gaussian variance $a \geq 0$ and Lévy measure ν, and $\Delta(S) = C_{\mathrm{ind}}\beta$. Notice that in (4.5), the Gaussian component a contributes 0 to β (as a limit of $N \to \infty$ summands of variance a/N whose third-moment sum $O(N^{-1/2})$ vanishes), while the jump measure ν contributes the limiting rare-summand sum $\int |y|^3\,\nu(dy)$. Even in this enlarged compactification, Bell [2, Theorems 1.1 and 1.2] prove that every three-point maximizing representation satisfies $a = 0$, $\nu = 0$, and $\mathrm{diam}(\mathrm{supp}\, X_i) \to 0$; if infinitely many X_i are nonconstant, then S is nonatomic, and under $\beta \leq 18481/25000$ the variances satisfy

$$\sum_{i\geq 1} \mathrm{Var}(X_i)\,\mathrm{diam}(\mathrm{supp}\, X_i)^{-\gamma} < \infty$$

for all $0 < \gamma < \sqrt{5} - 1$ (and all $0 < \gamma < \sqrt{21} - 3$ when all but finitely many summands are two-point), ruling out two-sided $C^{1,1}$ regularity of F_S at a maximizing threshold.

4.4 Proved classifications of maximizers and the binomial phase law

In Manuscripts 5 and 6 [1, 6], the structural classification question (Question 2) is solved completely in three regimes, and the asymptotic behavior of the optimal zero padding in $P_n = \max_{1\leq m\leq n} B_m$ is determined:

(1) *Common-diameter two-point arrays at every horizon $n \geq 1$* [1, Theorem 1.1]: Let C_n^{com} denote the maximum of $\mathcal{C}(X_1, \dots, X_n)$ over n-summand two-point arrays whose nondegenerate members share a common support diameter $h > 0$ (permitting zero summands). Then for every $n \geq 1$,

$$C_n^{\mathrm{com}} = P_n = \max_{1 \leq m \leq n} B_m, \tag{4.6}$$

and every common-diameter array attaining C_n^{com} consists of $m \in \{1, \dots, n\}$ identically distributed nondegenerate two-point summands and $n - m$ zero summands. The proof expresses the convolution probabilities as affine functions of the elementary symmetric polynomials of the success probabilities $(p_1, \dots, p_m)$ and combines Hoeffding's fiber theorem [17] with certified negative-curvature enclosures along constant-mean probability fibers.

(2) *Two two-point summands with arbitrary diameters ($n = 2$)* [1, Theorem 1.2]: Over arrays of at most two independent two-point summands with arbitrary support diameters $h_1, h_2 > 0$, the maximum of $\mathcal{C}(X_1, X_2)$ equals $C_1 = B_1 \approx 0.3703515235$, and every attaining array has exactly one nonconstant summand (so $C_2^{\mathrm{bin}} = P_2 = B_1$). If both summands are required to have strictly positive variance, the supremum remains C_1 and is not attained.

(3) *Strict local improvement under common-diameter replacement* [1, Theorem 1.3]: Fix $c_0 > 0$. There exist constants $A_0, \sigma_0, \eta_0, \kappa > 0$, depending only on c_0, with the following property. Let $X_i = g_i(B_i - p_i)$ $(1 \leq i \leq n)$ be independent nondegenerate two-point summands with $g_i > 0$ and $0 < p_i < 1$, and set

$$a_i = p_i(1 - p_i), \qquad A = \sum_{i=1}^{n} a_i, \qquad h_0 = \frac{1}{A} \sum_{i=1}^{n} a_i g_i, \qquad e_i = \frac{g_i}{h_0} - 1,$$

$$\sigma = \left(\sum_{i=1}^{n} a_i e_i^2 \right)^{1/2}, \qquad \eta = \max_{1 \leq i \leq n} |e_i|.$$

If $A \geq A_0$, $0 < \sigma \leq \sigma_0$, $\eta \leq \eta_0$, and $\mathcal{C}(X_1, \dots, X_n) \geq c_0$, then replacing all diameters g_i by their variance-weighted mean h_0 while keeping $(p_1, \dots, p_n)$ fixed strictly increases the Berry–Esseen ratio:

$$\mathcal{C}(B_1 - p_1, \dots, B_n - p_n) - \mathcal{C}(X_1, \dots, X_n) \geq \kappa \frac{\sigma^2}{\sigma + \eta} > 0. \tag{4.7}$$

If in addition all p_i lie in a fixed compact subset $J \subset (0, 1)$, the gain is at least $\kappa_J \sigma$, whereas when rare probabilities $p_i \to 0$ are permitted the rate $\sigma^2/(\sigma + \eta)$ is sharp. (Note that the strict positivity hypothesis $0 < \sigma$ is essential, since $\sigma = 0$ means $g_1 = \cdots = g_n = h_0$ where the two sides of (4.7) are equal.)

(4) *Asymptotic lattice structure and exact classification for small Lyapunov ratio* [6, Theorems 1.1 and 1.2]: First, along any sequence of finite independent arrays with $L_n \to 0$ and $\mathcal{C}_n \to C_{\mathrm{E}}$, there exist lattice spacings $h_n \sim R_n / \big((p_{\mathrm{E}}^2 + q_{\mathrm{E}}^2) V_n\big)$ and centered laws $X_{n,i}^{\mathrm{lat}}$ supported on cosets of $h_n \mathbb{Z}$ such that

$$\sum_i W_2^2(X_{n,i}, X_{n,i}^{\mathrm{lat}}) = o(h_n^2), \qquad \sum_i W_3^3(X_{n,i}, X_{n,i}^{\mathrm{lat}}) = o(h_n^3),$$

while $\mathcal{B}(X_n) \to 1$ and a core of $m_n \sim (p_{\mathrm{E}}^2 + q_{\mathrm{E}}^2)^2 / (p_{\mathrm{E}} q_{\mathrm{E}} L_n^2)$ summands carries $1 - o(1)$ of V_n and R_n and is W_3-close to scaled Esseen two-point laws. Second, there exists an absolute constant $\varepsilon_0 > 0$ (established by compactness and not effective) such that *every* independent

array $(X_1, \dots, X_n)$ attaining the unrestricted n-summand maximum C_n with $L < \varepsilon_0$ consists of $m \in \{1, \dots, n\}$ identically distributed nondegenerate two-point summands and $n - m$ zero summands. To eliminate arbitrarily small rounding residuals $T = \sum_i Z_i$ modulo $h_n\mathbb{Z}$ that cannot be ruled out by asymptotic $o(h_n^2)$ bounds alone, the proof establishes a relative conditional first-moment bound $\mathbb{E}(|T| \mid K = k) \geq c\,\mathbb{E}|T|$ valid for arbitrarily small $\mathbb{E}|T| > 0$, proving that exact lattice rounding strictly increases $\mathcal{C}$ whenever $T \neq 0$; discrete Green-function concavity on $h_n\mathbb{Z}$ then reduces each summand to two adjacent lattice points, where (4.6) completes the classification.

(5) *Asymptotic phase expansion of B_m and optimal zero summands* [1, Theorem 4.1]: A uniform seventh-order Edgeworth–Euler–Maclaurin expansion of $R_m(p)$ around $p = p_{\mathrm{E}}$ shows that as $m \to \infty$,

$$B_m = C_{\mathrm{E}} - \frac{Q_*}{m} - \frac{H_* + \alpha \operatorname{dist}(mp_{\mathrm{E}} - \delta_0, \mathbb{Z})^2}{m^2} + O(m^{-3}), \tag{4.8}$$

where H_* is an explicit constant independent of m, and

$$Q_* = \frac{\phi(0)(867 + 287\sqrt{10})}{25920}, \qquad \alpha = \frac{\phi(0)(60 + 19\sqrt{10})}{30}, \qquad \delta_0 = \frac{2590 + 577\sqrt{10}}{12960}. \tag{4.9}$$

Consequently, when comparing $m = n - j$ active summands ($j \geq 0$ bounded) against n, choosing j zero summands minimizes the second-order loss

$$\eta j + \operatorname{dist}(\theta_n + jp_{\mathrm{E}}, \mathbb{Z})^2, \qquad \eta = \frac{Q_*}{\alpha} = \frac{2510 - 747\sqrt{10}}{8640} \approx 0.017104003, \tag{4.10}$$

where $\theta_n = \operatorname{round}(np_{\mathrm{E}} - \delta_0) - (np_{\mathrm{E}} - \delta_0) \in [-1/2, 1/2]$. Because 5η strictly exceeds the maximum over $\theta \in [-1/2, 1/2]$ of the lower envelope of the first five parabolas $j \in \{0, 1, 2, 3, 4\}$, at most four zero summands are ever optimal for sufficiently large n. Extending the expansion to $O(m^{-3})$ and using a quadratic-field norm bound in $\mathbb{Q}(\sqrt{10})$ to rule out exact ties between distinct branches $j \neq j'$ proves that for all sufficiently large n:

(a) the maximizing active count m_n in $P_n = \max_{1 \leq m \leq n} B_m$ is unique;

(b) the number of zero summands $j_n = n - m_n$ belongs to $\{0, 1, 2, 3, 4\}$, and by Weyl equidistribution of the irrational rotation $np_{\mathrm{E}} \bmod 1$ the asymptotic frequencies of $j_n = 0, 1, 2, 3, 4$ are respectively

$$p_{\mathrm{E}} - \omega, \quad p_{\mathrm{E}} - \omega, \quad \sqrt{10} - 3, \quad \omega, \quad \omega, \tag{4.11}$$

where $\omega = (7773 - 2447\sqrt{10})/8640 \approx 0.00404464$;

(c) the gaps between successive record active counts m (those m for which $B_m > P_{m-1}$) belong to $\{2, 3, 5\}$.

5 Computational evidence and certified local obstructions

5.1 Scope of the stored computations

To interpret the numerical evidence in this repository accurately, one must distinguish three different kinds of computation:

(1) *High-precision floating-point evaluations in low-dimensional parametric families:* The file `data/empirical-observations.json` records deterministic evaluations of the one-variable

binomial ratio $R_m(p)$ for $1 \le m \le 3000$ and of finite binomial-plus-compound-Poisson families. These computations evaluate explicit closed-form or series expressions at high precision within specified parametric families; they are neither interval root-count certificates nor searches over arbitrary n-summand distributions.

(2) *Rigorous interval arithmetic certificates:* The certificate file `certificates/observations/local-trap.json`, verified by `verify/observations/local_trap.py` in 320-bit Arb ball arithmetic [18, 19] (alongside the interval certificates for Manuscripts 1, 2, and 5 under `certificates/`), provides outward-rounded computer-assisted mathematical proofs of specific inequalities, root enclosures, and Hessian sign conditions.

(3) *Unrestricted multi-summand numerical optimization:* No general numerical search over unrestricted multi-atom or arbitrary-diameter n-summand arrays is included in the verified repository records, and no claim in this paper or the companion manuscripts relies on uncertified exploratory searches.

5.2 High-precision binomial evaluations and zero padding through $m = 3000$

The dataset `data/empirical-observations.json` evaluates the i.i.d. binomial maxima $B_m = \max_{0<p<1} R_m(p)$ and the padded binomial envelope $P_n = \max_{1\le m\le n} B_m$ for $1 \le m \le 3000$ in two regimes:

- *Exhaustive threshold evaluation ($1 \le m \le 100$):* For each $m \in \{1, \dots, 100\}$, a uniform grid of 4096 probabilities in $(0, 1/2]$ (which covers $(0, 1)$ by reflection $p \leftrightarrow 1 - p$) evaluates every threshold index $0 \le k \le m$ and both one-sided CDF branches in (1.5). Visible local maxima are bracketed and refined by golden-section and stationary-root iteration to 60 decimal places, with stationarity residuals at most 2.61×10^{-59}.

- *Central-threshold scan ($101 \le m \le 3000$):* For each $m \in \{101, \dots, 3000\}$, the seven central threshold indices k nearest mp are evaluated and bracketed at cell endpoints. As a consistency check, for $101 \le m \le 450$ a denser control scan evaluates all central threshold cells, both CDF branches, and 17-point sub-meshes per cell with zero discrepancies; for ratios exceeding 0.4045, the tail bounds of Schulz [24] rigorously exclude noncentral thresholds.

Table 1 displays representative values of P_n and the optimal zero padding $j_n = n - m_n$. Several features merit emphasis:

(1) Across the entire scan $1 \le n \le 3000$, the observed number of zero summands $j_n = n - m_n$ takes only the five values $\{0, 1, 2, 3, 4\}$ proved asymptotically in Bell [1, Theorem 4.1]. At small horizons, successive record active counts m (those m with $B_m > P_{m-1}$) include consecutive integers with gap 1 (such as $m = 5, 6$, $m = 17, 18$, and $m = 29, 30$) before settling into the asymptotic gap set $\{2, 3, 5\}$ proved for large n in Bell [1, Theorem 4.1].

(2) The horizons $n = 412$ and $n = 413$ are the first values of n at which three and four zero summands occur: both retain the active count $m = 409$, whose 60-digit refined ratio compares with B_{412} and B_{413} as

$$\begin{aligned} B_{409} &= 0.409665276377194309 86\ldots, \\ B_{412} &= 0.409665275927097299 90\ldots \quad (B_{409} - B_{412} \approx 4.501 \times 10^{-10}), \\ B_{413} &= 0.409664784826714559 82\ldots \quad (B_{409} - B_{413} \approx 4.916 \times 10^{-7}). \end{aligned}$$

Table 1: Selected values of the padded binomial envelope $P_n = \max_{1\le m\le n} B_m = B_{m_n}$ from `data/empirical-observations.json`, showing the maximizing active count m_n, the number of zero summands $j_n = n - m_n$, the optimal success probability $p < \frac{1}{2}$, the normalized support diameter $h = 1/\sqrt{m_n p(1-p)}$, the Lyapunov ratio L, and the ratio P_n. Values for $m_n \le 100$ and $m_n \in \{409, 412, 413\}$ are rounded from 60-digit stationary-root refinements; $n = 414$ and $n = 3000$ are binary64 central-cell evaluations.

n	m_n	$j_n = n - m_n$	Probability p	Diameter h	Lyapunov L	Ratio $P_n = B_{m_n}$
1	1	0	0.390958631281333	2.049327	1.073396	0.370351523474379
2	1	1	0.390958631281333	2.049327	1.073396	0.370351523474379
3	3	0	0.431607387750433	1.165657	0.593733	0.398156326353304
4	3	1	0.431607387750433	1.165657	0.593733	0.398156326353304
10	10	0	0.429865759681911	0.638771	0.325669	0.406507736905768
12	10	2	0.429865759681911	0.638771	0.325669	0.406507736905768
50	49	1	0.414895409236786	0.289945	0.149173	0.409145180658100
100	99	1	0.417421147624576	0.203806	0.104683	0.409451697372009
409	409	0	0.418879394858355	0.100221	0.051430	0.409665276377194
412	409	3	0.418879394858355	0.100221	0.051430	0.409665276377194
413	409	4	0.418879394858355	0.100221	0.051430	0.409665276377194
414	414	0	0.418652506346914	0.099622	0.051129	0.409666049802979
3000	2999	1	0.418913502013957	0.037011	0.018992	0.409723068705426

(3) Comparing the empirical padding j_n with the leading quadratic phase rule (4.10) illustrates the role of the $O(m^{-3})$ error term in (4.8) near the narrow phase windows of width $\omega \approx 0.004045$ where $j \in \{3, 4\}$. Across the transition block $n = 409, 410, 411, 412, 413, 414$ (where $j_n = 0, 1, 2, 3, 4, 0$, respectively) and at $n = 2957, 2958$ (where $j_n = 3, 4$), the leading phase formula (4.10) already agrees with the observed padding j_n. At smaller or borderline horizons such as $n \in \{147, 148\}$, $n \in \{677, 678\}$, and $n \in \{1780, 1781\}$, the $O(1)$ leading phase formula lies near the boundary of the $\{3, 4\}$ window, and including the finite-m probability drift shifts the maximizer to $j_n \in \{1, 2\}$ (for $n = 147, 148$ and $1780, 1781$) or $j_n \in \{0, 1\}$ (for $n = 677, 678$), matching the observed active counts.

5.3 A certified strict local maximizer with unequal diameters

Why does the local diameter-equalization theorem (Theorem 1.3 of 1, recalled in (4.7)) require the diameter spread σ and maximum relative deviation η to be small? When all nondegenerate two-point summands share a common diameter h, the 2^m atoms of the convolution sum $S = \sum_{i=1}^m h(B_i - p_i)$ coalesce into $m + 1$ lattice points. Perturbing h_i splits the colliding atoms at the maximizing threshold and produces a downward directional cusp in $\mathcal{C}$. However, atom collisions also occur when unequal diameters have *rational ratios*, and such collisions can lock unequal diameters into a strict local maximum.

Consider $n = 4$ independent two-point summands $X_i = h_i(B_i - p_i)$ with support diameters

$$h = (1, 1, 1, 2), \tag{5.1}$$

upper atom probabilities $(p_1, p_2, p_3, p_4) = (p, p, p, r)$ near

$$p = 0.57275160060602450694769280 8\ldots, \qquad r = 0.33747296608998501783486870 4\ldots, \tag{5.2}$$

and raw threshold $k = 2$ on the closed branch $\mathbb{P}(\sum_{i=1}^4 h_i B_i \le 2) - \Phi(z_2)$, where $z_2 = (2 - 3p - 2r)/\sqrt{3p(1-p) + 4r(1-r)}$. At the raw sum value $\sum_{i=1}^4 h_i B_i = 2$, four binary configurations

$v = (B_1, B_2, B_3, B_4) \in \{0,1\}^4$ collide:

$$v^{(1)} = (1,1,0,0), \qquad v^{(2)} = (1,0,1,0), \qquad v^{(3)} = (0,1,1,0), \qquad v^{(4)} = (0,0,0,1). \tag{5.3}$$

These four vectors satisfy $v^{(j)} \cdot h = 2$ and form the vertices of a nondegenerate three-dimensional tetrahedron in the hyperplane $\{v \in \mathbb{R}^4 : v \cdot h = 2\}$.

Using 320-bit Arb ball arithmetic on the closed box of radius 10^{-28} around (5.2), the verification script `verify/observations/local_trap.py` checks the interval certificate `certificates/observations/local-trap.json` and establishes the following theorem:

Theorem 5.1 (Certified unequal-diameter strict local maximizer)**.** *In the four-summand two-point family $X_i = h_i(B_i - p_i)$ with $h = (1,1,1,2)$, the closed box of radius 10^{-28} around (p, r) in (5.2) contains a unique point (p_*, r_*) at which the probabilities $p^{(0)} = (p_*, p_*, p_*, r_*)$ are stationary for the closed-branch ratio at raw threshold $k = 2$. At $(h; p^{(0)})$:*

(i) The Kolmogorov distance $\Delta(S)$ is uniquely attained at the raw threshold $k = 2$ on the closed branch, and the Berry–Esseen ratio satisfies

$$\mathcal{C} \in 0.28248894967964153979405802 \pm 5.54 \times 10^{-27}. \tag{5.4}$$

(ii) The 4×4 Hessian $H_p = \nabla_p^2 \mathcal{C}(h; p^{(0)})$ with respect to the probabilities (p_1, p_2, p_3, p_4) at fixed diameters $h = (1,1,1,2)$ is strictly negative definite.

(iii) Under any perturbation $h \mapsto h + d$ ($d \in \mathbb{R}^4$ small) of the support diameters, the four colliding configurations (5.3) move to the raw sum values $v^{(j)} \cdot (h + d) = 2 + v^{(j)} \cdot d$ ($1 \le j \le 4$). Any candidate threshold near 2 that retains all four colliding configurations in the closed ray has the form $2 + v^{(j_*)} \cdot d$ for an active index $j_* \in \{1,2,3,4\}$ maximizing $v^{(j)} \cdot d$ (any lower threshold omits at least one colliding configuration of positive probability). Along each smooth branch function $f_j(h + d, p^{(0)} + \delta p)$ corresponding to the threshold $v^{(j)} \cdot (h + d) = 2 + v^{(j)} \cdot d$, the diameter gradient at $(0, p^{(0)})$ is $c_1(\beta - v^{(j)})$ with $c_1 > 0$, where $\beta = \sum_{j=1}^4 w_j v^{(j)}$ lies in the strict interior of the tetrahedron $\operatorname{conv}\{v^{(1)}, v^{(2)}, v^{(3)}, v^{(4)}\}$ with barycentric weights $w_1 = w_2 = w_3 \in 0.31153 \pm 10^{-5}$ and $w_4 \in 0.06541 \pm 10^{-5}$. Consequently, for every $d = d_\perp$ orthogonal to $h = (1,1,1,2)$ and any active index j_* maximizing $v^{(j)} \cdot d_\perp$,

$$\nabla_d f_{j_*}(0, p^{(0)}) \cdot d_\perp = c_1 \Big(\beta \cdot d_\perp - \max_{1 \le j \le 4} v^{(j)} \cdot d_\perp \Big) \le -c_2 \|d_\perp\|, \tag{5.5}$$

with $c_2 > 0$. Since $f_j(0, p) = \mathcal{C}(h; p)$ has vanishing probability gradient $\nabla_p \mathcal{C}(h; p^{(0)}) = 0$ and negative-definite probability Hessian $H_p \preceq -c_H I < 0$, a second-order Taylor expansion of f_{j_*} around $(0, p^{(0)})$ with $a = \|d_\perp\|$ and $b = \|\delta p\|$ gives

$$\mathcal{C}(h + d_\perp; p^{(0)} + \delta p) - \mathcal{C}(h; p^{(0)}) \le -c_2 a - \frac{c_H}{2} b^2 + O(ab + a^2 + b^3) < 0$$

for all sufficiently small $(d_\perp, \delta p) \ne (0,0)$, because $ab \le b(a + b^2) = o(a + b^2)$ and $a^2 + b^3 = o(a + b^2)$ as $a, b \to 0$.

(iv) For both the unit-diameter summands ($i = 1,2,3$) and the diameter-2 summand ($i = 4$), the single-coordinate variational contact cubic majorants hold globally on $\mathbb{R}$, with certified strictly positive gaps at all non-contact breakpoints of the deleted-sum distribution functions.

Theorem 5.1 shows that $(h; p^{(0)})$ is a strict local maximizer of $\mathcal{C}$ under joint perturbations of (h_1, h_2, h_3, h_4) and (p_1, p_2, p_3, p_4) modulo common scaling, and satisfies all single-coordinate variational contact inequalities on $\mathbb{R}$. Yet its Berry–Esseen ratio $0.282489\ldots$ is substantially smaller than the ratio $\sqrt{2}/4 \approx 0.353553$ of two symmetric Bernoulli summands and two zeros (evaluated at threshold 0), and far below the four-summand common-diameter maximum $P_4 = B_3 \approx 0.398156$. This example demonstrates that local stationarity and single-coordinate contact conditions cannot by themselves force support diameters to be equal when $n \geq 3$.

5.4 Binomial sums with a compound-Poisson jump component

The dataset `data/empirical-observations.json` (under `mixed_core_cp`) also records numerical comparisons for centered sums of the form

$$S_{\mathrm{raw}} = \big(\mathrm{Bin}(M, p) - Mp\big) + g\big((N_+ - \ell_+) - (N_- - \ell_-)\big), \tag{5.6}$$

where $N_\pm \sim \mathrm{Pois}(\ell_\pm)$ are independent of $\mathrm{Bin}(M, p)$, across 18 parameter profiles with $M \in \{1, 2, 3, 5, 10, 20\}$, jump step $g \in \{1/2, 1, 2\}$, $p \in [0.05, 0.5]$, and total intensity $10^{-6} \leq \ell_+ + \ell_- \leq 12$. In this experiment, the variance and the third-moment functional are taken to be

$$V = Mp(1-p) + g^2(\ell_+ + \ell_-), \qquad \beta = Mp(1-p)(p^2 + (1-p)^2) + g^3(\ell_+ + \ell_-). \tag{5.7}$$

As emphasized in Remark 2.4 and Section 4.3, the term $g^3(\ell_+ + \ell_-)$ in (5.7) is the Lévy jump integral $\int_{\mathbb{R}\setminus\{0\}} |y|^3 \, \nu(dy)$ representing the limiting sum of third absolute moments of arbitrarily many rare two-point summands of step sizes $\pm g$ in the all-horizon compactification (4.5). It is not the third absolute moment of a single centered Poisson variable, and by Remark 2.4(2) it is strictly smaller than the minimum third-moment sum $\mathcal{M}_{n-M}$ of any finite $(n-M)$-summand independent factorization of the compound-Poisson component.

Even when the compound-Poisson component is charged only this smaller limiting third-moment cost $g^3(\ell_+ + \ell_-)$, across all 18 tested profiles every candidate with $\ell_+ + \ell_- > 0$ has ratio $\Delta(S_{\mathrm{raw}})/(\beta/V^{3/2})$ strictly smaller than the pure binomial ratio $R_M(p)$ ($\ell_+ = \ell_- = 0$) at the same probability p. This provides a finite-M numerical illustration of the theorem $\nu = 0$ proved for compactified global maximizers in Bell [2, Theorem 1.1].

6 What remains open

6.1 The finite-n conjectures

Combining the theoretical results of Sections 2–4 and the numerical evidence of Section 5, we can state the three open finite-n conjectures corresponding to Questions 1–3.

Conjecture 6.1 (Finite-n value conjecture)**.** For every integer $n \geq 1$,

$$C_n = P_n = \max_{1 \leq m \leq n} B_m.$$

In particular, $C_{\mathrm{ind}} = \sup_{n \geq 1} C_n = \lim_{m \to \infty} B_m = C_{\mathrm{E}}$.

Conjecture 6.2 (Finite-n extremizer classification conjecture)**.** For every integer $n \geq 1$, every n-summand independent array $(X_1, \ldots, X_n)$ attaining C_n consists of $m \in \{1, \ldots, n\}$ identically distributed nondegenerate two-point random variables and $n - m$ zero variables, up to a common nonzero scale factor, sign reflection, and permutation of coordinates.

Table 2: Summary of what is currently proved toward Conjectures 6.1–6.3.

Regime	Proved result	Reference
$n = 1$, unrestricted	$C_1 = B_1$; unique two-point maximizer	Bentkus and Kirša [8]
$n = 2$, two-point	$C_{2,\mathrm{bin}} = P_2 = B_1$; one summand zero	Bell [1, Thm. 1.2]
$n = 2$, unrestricted	$C_2 < 49/120 < C_{\mathrm{E}}$ ($C_2 = B_1$ open)	Bell [3, Thm. 1.1]
$n = 3$, two-point	$\Delta \le C_{\mathrm{E}} L$ ($\sup \mathcal{C} = B_3$ open)	Bell [3, Thm. 1.2(i)]
All $n \ge 1$, common-diam. two-point	$C_n^{\mathrm{com}} = P_n$; maximizers padded i.i.d.	Bell [1, Thm. 1.1]
All $n \ge 1$, $L \le \exp(-\exp(97))$	$\Delta \le C_{\mathrm{E}} L$	Bell [5, Thm. 1.1]
All $n \ge 1$, maximizers with $L < \varepsilon_0$	$C_n = P_n$; maximizers padded i.i.d.	Bell [6, Thm. 1.2]
All $n \ge 1$, unrestricted	$C_{\mathrm{ind}} < 0.44988$; ≤ 4 pts ($\le$ 3-pt rep.)	Bell [2, 4]
Large n, binomial envelope P_n	Unique m_n; $n - m_n \in \{0, 1, 2, 3, 4\}$	Bell [1, Thm. 4.1]

Conjecture 6.3 (Uniqueness of the binomial maximizer)**.** (i) For every integer $m \ge 1$, the function $p \mapsto R_m(p)$ in (1.5) attains its maximum B_m on $(0, 1/2]$ at a unique probability $p_m \in (0, 1/2)$.

(ii) For every integer $n \ge 1$, the padded binomial envelope $P_n = \max_{1 \le m \le n} B_m$ is attained at a unique active count $m_n \in \{1, \dots, n\}$.

Table 2 summarizes the proved cases. Note that Conjecture 6.2 at a given n immediately implies Conjecture 6.1 at that n.

6.2 Overall proof architecture and the missing global implication

The results of this paper and the companion manuscripts assemble the machinery for two complementary approaches to Conjecture 6.2, and make it transparent which implications are established and which remain open in each:

Route 1. (Reduction to sum-law selection via exact-fiber factorization) In Section 2, we separated the optimization of $\mathcal{C}(X_1, \dots, X_n)$ into two stages:

$$C_n = \sup_{\mu} \mathcal{Q}_n(\mu), \qquad \mathcal{Q}_n(\mu) = \frac{V(\mu)^{3/2} \Delta(\mu)}{\mathcal{M}_n(\mu)},$$

where $\mu = \mathrm{Law}(S)$ is the distribution of the sum and $\mathcal{M}_n(\mu)$ is the minimum third-moment sum over all n-summand independent factorizations of μ. By Theorem 2.3(ii), once μ is known to be a centered binomial law $\mathrm{Law}(g(\mathrm{Bin}(m, p) - mp))$ of order $m \le n$, minimizing $\sum_{i=1}^n \mathbb{E}|X_i|^3$ on the fiber $\mathcal{F}_n(\mu)$ *automatically* forces the summands $(X_1, \dots, X_n)$ to consist of m identically distributed two-point variables and $n - m$ zeros. More generally, by Theorem 2.3(i), if μ is any Poisson-binomial law of order $k \le n$, fiber minimization forces every nonconstant summand to be a two-point variable of common diameter g, and Theorem 1.1 of Bell [1] then forces their probabilities p_j to be equal.

What remains unproved in Route 1: Proving that the sum distribution $\mu = \mathrm{Law}(S)$ of every global maximizer of C_n is a Poisson-binomial (or binomial) law of order at most n. Minimizing the factorization cost $\mathcal{M}_n(\mu)$ for a fixed μ operates within a single convolution fiber $\mathcal{F}_n(\mu)$; it does not compare $\mathcal{Q}_n(\mu)$ across different sum distributions μ.

Route 2. (Reduction to characteristic-function coherence) In Section 3, Theorem 3.3 showed that for any independent array $(X_1, \dots, X_n)$, the moment-determined span $\widehat{h} = (R+D)/V$ defines a nonnegative characteristic-function deficit $\mathcal{J} = 1 - |\mathbb{E}e^{2\pi \mathrm{i} S/\widehat{h}}|^2 \geq 0$ such that $\mathcal{J} = 0$ if and only if $(X_1, \dots, X_n)$ is a padded i.i.d. two-point array. Notice that $\mathcal{J} = 0$ combines two properties simultaneously: it places the support of every summand on a coset of $\widehat{h}\mathbb{Z}$ and saturates the separated-support moment inequality of Theorem 3.1 at the same span $\widehat{h}$. (By contrast, merely knowing that the summands of a fixed-n maximizer are supported on cosets of some common lattice $h\mathbb{Z}$ does not by itself reduce them to two-point laws without an additional step: the discrete Green-function concavity argument of 6, Section 9 requires $L < \varepsilon_0$ to supply asymptotic local limit estimates for the deleted sum, while Theorem 3.1 requires equality in the moment bound (3.5) in addition to support separation h.)

What remains unproved in Route 2: Proving that global optimality for $\mathcal{C}(X_1, \dots, X_n)$ forces $\mathcal{J} = 0$. As emphasized in Remark 3.4, Theorem 3.3 is an equality characterization for $\mathcal{J} = 0$, not an inequality bounding $\mathcal{C}$ by a decreasing function of $\mathcal{J}$. Moreover, while Bell [2, Theorem 3.2] guarantees that every fixed-n maximizer has at most four support points per summand and admits an equally maximizing replacement with at most three support points per summand (which is convolution-indecomposable when at least one zero slot is present), proving the value conjecture $C_n = P_n$ by this route suffices if at least one three-point maximizing representative for each n satisfies $\mathcal{J} = 0$, whereas proving the full classification conjecture (Conjecture 6.2) requires showing that every three-point maximizing representative satisfies $\mathcal{J} = 0$ and that no four-point flat-segment maximizer can exist.

6.3 Specific analytical obstacles in the remaining regimes

Three concrete analytical phenomena explain why extending the proved classifications to all $n \geq 2$ (for general summands) and all $n \geq 3$ (for two-point summands with arbitrary diameters) requires new global arguments:

(1) *Zero summands decouple the horizon n from the Lyapunov ratio L:* Theorem 1.2 of Bell [6] proves Conjectures 6.1 and 6.2 for any maximizer of C_n whose Lyapunov ratio satisfies $L < \varepsilon_0$. In the i.i.d. setting, $L_n = (p^2 + q^2)/\sqrt{npq} \to 0$ automatically as $n \to \infty$. In the independent setting, however, adjoining zero summands increases n without changing L: an n-summand array can have $L \geq \varepsilon_0$ for arbitrarily large n if all but a bounded number of summands have zero or negligible variance. Because $\varepsilon_0 > 0$ in Bell [6] is obtained by a compactness contradiction argument and is not effective, ruling out maximizers with $L \geq \varepsilon_0$ requires a finite-n classification at moderate Lyapunov ratios.

(2) *Atom collisions at rational diameter ratios create strict local traps:* For two-point summands with arbitrary support diameters $h = (h_1, \dots, h_n)$, Theorem 5.1 proves that at $n = 4$ the unequal-diameter configuration $h = (1, 1, 1, 2)$ is a strict local maximizer of $\mathcal{C}$ that satisfies every single-coordinate variational contact inequality globally on $\mathbb{R}$. Whenever a subset of diameters has rational ratios, multiple binary configurations can collide at the maximizing threshold and place the effective weight vector β in the interior of the collision polytope in (5.5), locking the diameters against all local perturbations. Thus any proof that $C_n^{\mathrm{bin}} = P_n$ for $n \geq 3$ must use a global replacement (capable of jumping across collision hyperplanes or comparing the multi-collision value against P_n) rather than local gradient ascent.

(3) *Dependence of single-summand replacements on the deleted-sum distribution:* When one modifies a single summand X_i while holding the remaining summands fixed, the numerator of $\mathcal{C}$ at a threshold x is $\left|\mathbb{E}F_{T_i}(x - X_i) - \Phi(x/\sqrt{V})\right|$, where F_{T_i} is the distribution function of the deleted sum $T_i = \sum_{j\neq i} X_j$. Unlike the third-moment sum $R = \sum_{i=1}^{n} \mathbb{E}|X_i|^3$, which is additive across coordinates, $\mathbb{E}F_{T_i}(x - X_i)$ couples X_i to the step locations and jump sizes of T_i. As Proposition 3.5 illustrates for zero-bias transport, and as Theorem 5.1 illustrates for unequal-diameter two-point arrays, a marginal replacement that does not account for the alignment of atoms with the threshold x and the deleted sum T_i can decrease $\mathcal{C}$ even when it moves X_i toward a two-point or equal-diameter law. Establishing an admissible global comparison that overcomes this coupling is the central open problem in the finite-n theory of the independent Berry–Esseen inequality.

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